University of Geneva
Faculty of Science
Master of Science in Computer Science

Master's Thesis
Personality Compatibility and Collaboration: An MBTI Approach to Actor Relationships

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Abstract

The Myers-Briggs Type Indicator (MBTI) is widely used to describe individuals' social interaction patterns, yet its actual influence on the structuring of social networks remains debated. This work investigates whether a link exists between an actor's MBTI personality type and their relationships. Based on a co-casting network constructed over the period 1990–2009, we computed a set of centrality metrics (degree, betweenness, closeness, PageRank, eigenvector, clustering coefficient, rich-club effect) to characterize actors' structural positions according to their MBTI type. Unsupervised K-Means clustering was then applied to the four continuous dimensions of the MBTI to search for the potential emergence of personality-based communities. The results reveal no significant structuring of the collaboration network by MBTI type: central positions and detected communities gather heterogeneous personality profiles, with no marked tendency toward either homophily or heterophily. These findings suggest that MBTI personality is not a decisive factor in large-scale professional collaboration network organization, contrary to what is sometimes observed in restricted, structured team contexts.

Keywords: data, MBTI, networks, link, collaboration, homophily, heterophily, centrality, K-Means, classification, actor, film, personality

Contents

1 Introduction

1.1 MBTI

The MBTI, an acronym for Myers-Briggs Type Indicator, is a personality assessment tool developed in the 20th century. It aims to better understand individual differences by classifying people's psychological preferences across four major dimensions:

- Extraversion (E) / Introversion (I)
- Sensing (S) / Intuition (N)
- Thinking (T) / Feeling (F)
- Judging (J) / Perceiving (P)

The functioning of the MBTI relies on how individuals perceive the world and make decisions. It identifies a personality type from sixteen possible combinations. This assessment does not measure skills or abilities, but rather natural preferences in daily functioning. Used in various contexts such as personal development, career guidance, or teamwork, it primarily represents a framework for interpreting human behavior.

1.2 Purpose/Motivations

Continuing this personality analysis, we seek to discover whether a potential link exists between individuals' personalities and those of the people they associate with. The objective is to understand whether our social relationships reflect, at least in part, our own way of thinking, feeling, and acting, or if, conversely, they rely more on profile complementarities.

To achieve this, we examine an actor network to observe potential trends in interpersonal relationships. The goal is to determine whether certain personalities tend to gravitate toward similar profiles sharing common characteristics or if they are more attracted to individuals with opposing operational styles. This approach will thus explore how psychological preferences can influence relational choices and the construction of social circles.

1.3 State of the Art

Social relationships have long been studied through the concept of homophily, defined by Lazarsfeld and Merton [7] as the tendency of individuals to establish ties with people who resemble them. Conversely, heterophily designates the attraction between individuals with different characteristics, emphasizing a logic of complementarity rather than similarity. This conceptual pair has been applied to numerous identity markers [8] before being extended to personality traits.

In the fields of psychology and network science [2, 3, 11, 5, 10, 4], several works have addressed the link between personality and the structuring of social relationships, with mixed results depending on the studied context.

Several studies conducted in organizational or educational contexts highlight homophilic tendencies. In software development, Gilal et al. [6] applied a complex network approach to MBTI data from software engineering students to identify the most "central" personality profiles in teammate choices. Their results show that types such as `INTJ` and `ESTP` are most frequently chosen as partners, whereas `ENTJ` and `INTP` stand out for high closeness centrality, suggesting an ability to work closely with other profiles. Conversely, types like `INFP`, `ISFP`, `INFJ`, and `ISTJ` appear poorly connected in the network, partially contradicting earlier literature attributing the quintessential programmer profile to `ISTJ`, illustrating the complexity and occasional inconsistency of results depending on the chosen measurement angle.

Other research highlights the importance of complementarity over similarity, particularly in contexts where roles are differentiated, such as software team work. Garousi and Tarhan [4] conducted an exploratory case study of 50 software engineering students divided into homogeneous or mixed groups based on the MBTI Introversion/Extraversion dimension. Their results show that personality composition has no significant effect on team cohesion, but mixed groups obtain better average project grades—particularly for students with lower academic averages—suggesting that personality diversity can compensate for technical skill gaps without harming social cohesion. This contrast underscores that the effect of personality homogeneity or heterogeneity strongly depends on the type of task performed by the group.

Another branch of literature, rather than focusing on ties between individuals, seeks to establish correspondences between personality types and aptitude for specific software life cycle roles. Capretz and Ahmed [3] proposed a mapping of the four MBTI dimensions to the five major phases of software development (analysis, design, coding, testing, maintenance) based on job postings and expected competencies. They concluded, for example, that `EF` profiles are preferable for system analysis, `NT` for design, `IST` for coding, `SJ` for testing, and `SP` for maintenance.

Varona and Capretz [10] deepened this approach by formalizing a "natural disposition coefficient," calculated as the ratio between rules satisfied by a given MBTI type and total rules identified for a role, based on software practitioners' reported preferences for tasks. This modeling produces a decision matrix supporting human resource allocation in software projects, yet relies on the implicit assumption that certain types are inherently better disposed for certain roles.

Thatcher and De la Cour [9] examined whether the effect of personality (MBTI) on small-group decision-making processes varied between face-to-face and computer-mediated communication. Contrary to the dominant literature hypothesis, extraversion (frequently invoked to explain participation differences) showed no significant effect, neither face-to-face nor computer-mediated. Only the `T/F` (Thinking/Feeling) dimension emerged robustly, with Feeling profiles obtaining higher scores in leadership, initiative, and interpersonal sensitivity, regardless of the medium. This result cautions against the central place often given to extraversion in personality-based social interaction studies.

The persistence of personality homophily when standard identity markers are suppressed represents a more recent research axis, directly relevant to studying large-scale actor networks. Ayyoubzadeh et al. [1] conducted a complex network analysis on over 288,000 messages exchanged among $6,076$ anonymous chat users, of whom 940 voluntarily provided MBTI type and gender. Results indicate a clear dominance of heterophilic interactions (89.3%), especially between cognitively complementary types (`NT` and `NF`), whereas personality homophily accounted for only 10.7%, concentrated mainly among intuitive-introverted types (`INTJ, INFP, INFJ`). The network exhibits scale-free properties with high-centrality hub nodes ensuring community cohesion, and relatively low modularity indicating weak community structure from personality. Notably, gender proved a much more robust homophily factor than MBTI type: female users showed a 55.6% gender homophily rate versus 23.4% for males, suggesting anonymity does not uniformly suppress all social grouping markers.

This result fits into a broader theoretical debate between two competing hypotheses: the homophily persistence hypothesis (personality signals persist under anonymity) and the heterophily amplification hypothesis (anonymity reduces standard social constraints and encourages exploration of different profiles). Studies cited therein also support weakened personality homophily in anonymous contexts, favoring other similarity forms.

Thus, although the influence of personality on social relationships is widely recognized, whether individuals prioritize similarity or complementarity remains open and strongly depends on:

- the type of task performed by the group,
- the degree of context structuring (project team with defined roles vs. free interaction),
- and the degree of anonymity in the studied environment.

It is important to note that many studies in this field have been conducted on specific populations, notably student groups or small professional teams. These contexts offer advantages of easy data collection and fine experimental control, but limit generalization to broader, more diverse populations. Furthermore, several research efforts focus on structured environments [9, 4, 6, 11] (work teams, project groups), where relations are constrained by common goals and predefined roles, potentially artificially reinforcing or masking personality effects that would manifest differently in a freer social framework. This issue justifies continuing investigations, notably through large-scale actor network analysis where large data volumes are available, to better understand spontaneous mechanisms driving interpersonal tie formation based on personality traits.

2 Methodology

2.1 Structure and Working Environment

This work can be divided into three parts:

- Actor/movie data processing
- MBTI data processing
- Network construction and analysis

Realized in Python 3.13, we used notebooks providing better visual feedback and structure for our work and results. Our three notebooks corresponding to the three parts are available on the Git repository: `https://github.com/DavidGabrielPapa/TheseDeMaster`

2.1.1 Actor/Movie Data Processing

The work for this part is found in the notebook: `dataset.ipynb`.

Several Python libraries were used across different steps:

Libraries	**Uses**
kagglehub, os	Online database reading and importing
pandas	Database manipulation and visualization
matplotlib.pyplot	Visuals for charts
json	Reading `.json` files containing movie casts
networkx	Network creation and characteristic analysis

Table 1: List of Python libraries and their uses

2.1.2 MBTI Data Processing

The work for this part is found in the notebook: `mbti.ipynb`.

As before, several Python libraries were used. Previously listed libraries were used similarly here without being re-listed:

Libraries	**Uses**
difflib	Uses method *get_close_matches()* to match 2 names
scipy.stats	Uses method *chi2_contingency()* to compute χ^2
numpy	Performs mathematical calculations
seaborn	Used to create heatmaps
unicodedata, re	Text modification and cleaning
collections	Used in Fuzzy-names

Table 2: List of Python libraries and their uses

2.1.3 Network Construction and Analysis

The work for this part is found in the notebook: `graph.ipynb`.

Previously listed libraries were used similarly here, listing only newly introduced ones:

Libraries	Uses
sklearn.cluster	Computes K-means and DBSCAN
sklearn.preprocessing	Centers and scales data
sklearn.metrics.pairwise	Uses method *euclidean_ distances*
itertools	Generates combinations from dimension lists
community	Computes communities using Louvain
kneed	Computes K-means elbow point

Table 3: List of Python libraries and their uses

2.2 Actor Data

Initially, we use a movie database (DB) also containing actor casts. These data originate from Kaggle. All manipulations and retrieved information are performed within `dataset.ipynb`.

To get a better overview of our database, we first analyze the distribution of movies by decade.

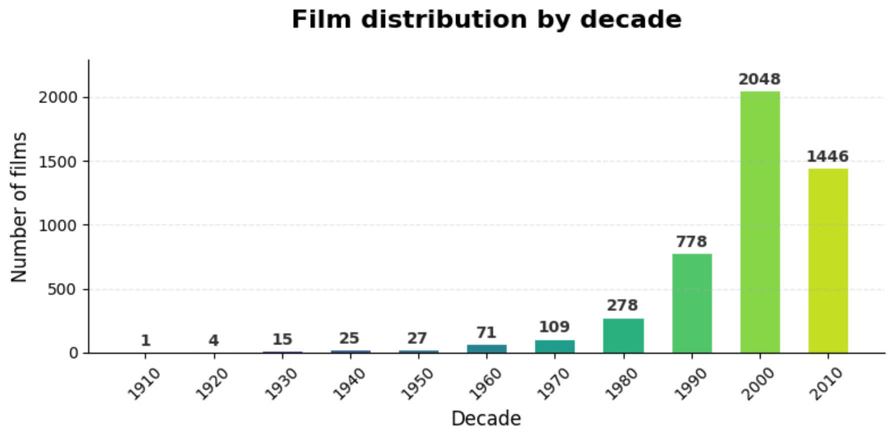

Figure 1: Movies by decade

We note that our DB contains mostly recent movies (post-1990), providing 2,826 movies for our research between 1990 and 2009. Working with actors, we count the unique actors contained and compare with the target period:

Description	Number	Ratio (%)
Number of unique actors (all database)	54 201	100%
Number of unique actors between 1990 and 2009	30 748	56.73%

Table 4: Number of actors

Actors being stored in `.json` format, we use Python's `json` library for easy access. Using the function **get_top_actors(cast_json, nbr)**, we define the number (nbr) of actors extracted per movie by importance order (if nbr=1, main actor only). If requested exceeds available, we return the maximum available. This serves for network construction.

2.3 MBTI Data

Personality information comes from a Kaggle database containing worldwide famous personalities.

Being comprehensive and containing unnecessary information, we reduce it by keeping relevant columns.

```
# Columns to keep
cols = ['name', 'category', 'subcategory', 'four_letter', 'letter_1',
    'letter_1_percentage', 'letter_2', 'letter_2_percentage', 'letter_3',
    'letter_3_percentage', 'letter_4', 'letter_4_percentage']

mbti = mbti[cols]
```

Next, we remove identical rows to prevent errors.

Description	Number of rows
Before cleaning	50 878
After cleaning	50 791

Table 5: Number of rows in MBTI DB

We analyze our database separating it three ways.

2.3.1 Entire Database

To identify potential trends for later study, we visualize personality profile distributions across our DB.

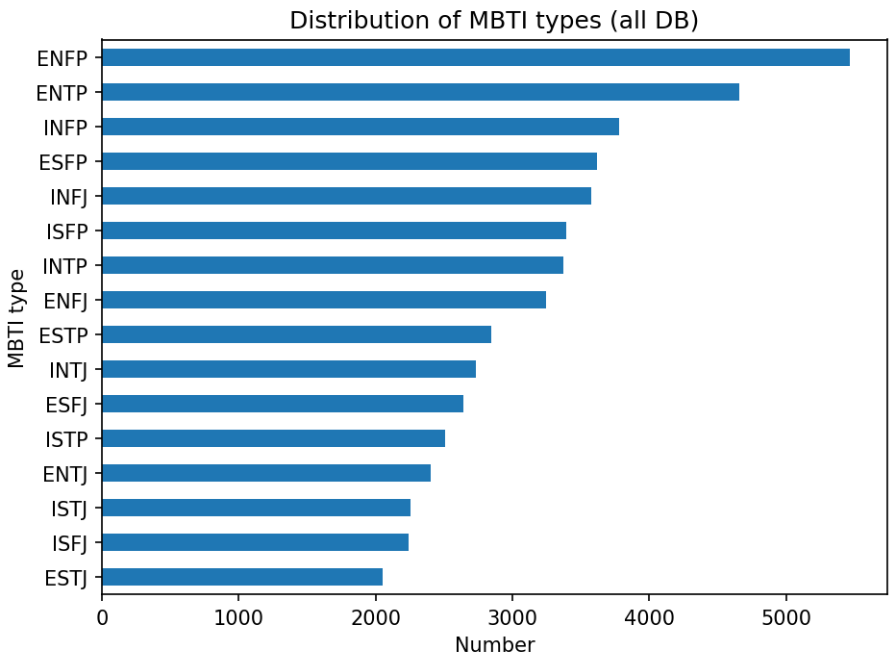

Figure 2: MBTI type distribution (all DB)

MBTI Type	Count	Percentage
ENFP	5 465	10.8%
ENTP	4 659	9.2%
INFP	3 776	7.4%
ESFP	3 619	7.1%
INFJ	3 578	7.0%
ISFP	3 392	6.7%
INTP	3 371	6.6%
ENFJ	3 247	6.4%
ESTP	2 847	5.6%
INTJ	2 732	5.4%
ESFJ	2 643	5.2%
ISTP	2 508	4.9%
ENTJ	2 402	4.7%
ISTJ	2 255	4.4%
ISFJ	2 243	4.4%
ESTJ	2 054	4.0%

Table 6: MBTI type distribution (entire DB)

After a global view, we observe a decreasing distribution without clear trends. We examine letter distribution for more precision:

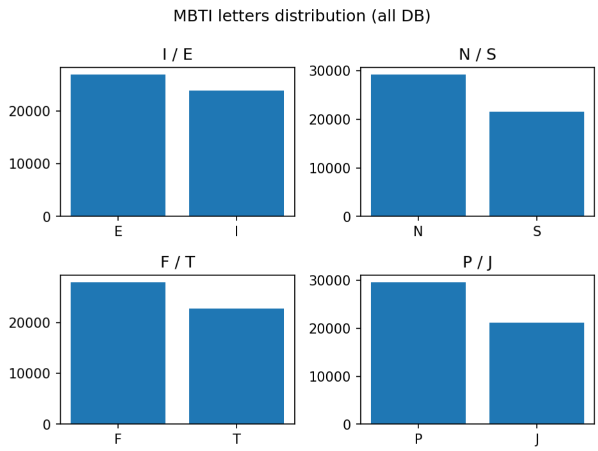

Figure 3: Letter distribution by position (all DB)

Position	Letter A	Letter B	Ratio
1st letter	E: 26 936	I: 23 855	53.0% - 47.0%
2nd letter	N: 29 230	S: 21 561	57.5% - 42.5%
3rd letter	F: 27 963	T: 22 828	55.1% - 44.9%
4th letter	P: 29 637	J: 21 154	58.4% - 41.6%

Table 7: Letter distribution by position (all DB)

To ensure dimensions are independent, we review their correlation matrix.

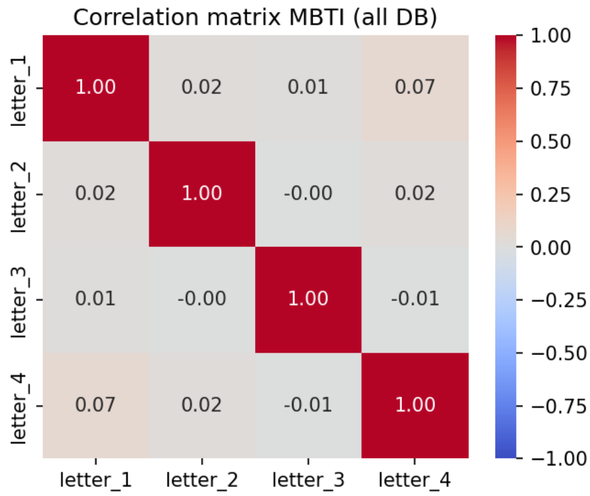

Figure 4: Correlation matrix (all DB)

Values being close to 0, no link is noted between dimensions. To verify, we compute the p-value testing variable independence. To support this, we compute Cramér's V measuring categorical association strength.

χ^2 Test	p-value	Cramér's V
letter_1 - letter_2	0.0000	0.0216
letter_1 - letter_3	0.0101	0.0114
letter_1 - letter_4	0.0000	0.0698
letter_2 - letter_3	0.6378	0.0021
letter_2 - letter_4	0.0001	0.0173
letter_3 - letter_4	0.2457	0.0052

Table 8: p-value and Cramér's V tests (entire DB)

Although some p-values are close to 0, all Cramér's V values are below 0.1, indicating no link between dimensions.

2.3.2 All Actors in Database

To be more precise, we examine individuals recorded as "actor" using the "subcategory" field:

Description	Number
Entire database	50 791
All DB "actors"	6 914

Table 9: Number of actors in MBTI DB

We note a sharp decrease in data quantity. To ensure relevant and coherent results, we check distribution by MBTI profile:

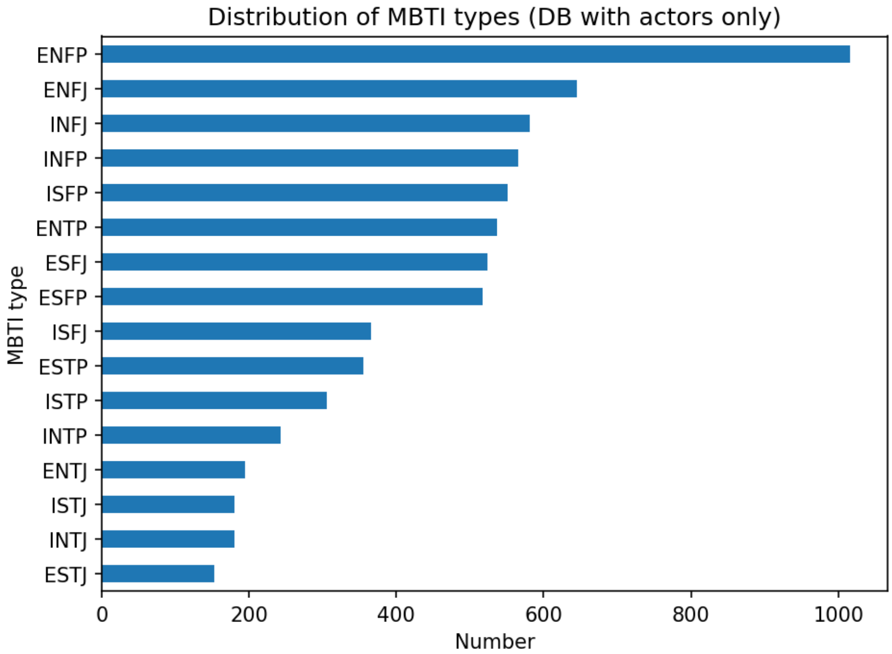

Figure 5: MBTI type distribution (for actors)

MBTI Type	Count	Percentage
ENFP	1 016	14.7%
ENFJ	645	9.3%
INFJ	581	8.4%
INFP	565	8.2%
ISFP	551	8.0%
ENTP	537	7.8%
ESFJ	524	7.6%
ESFP	517	7.5%
ISFJ	366	5.3%
ESTP	355	5.1%
ISTP	306	4.4%
INTP	243	3.5%
ENTJ	195	2.8%
INTJ	180	2.6%
ISTJ	180	2.6%
ESTJ	153	2.2%

Table 10: MBTI type distribution (for actors)

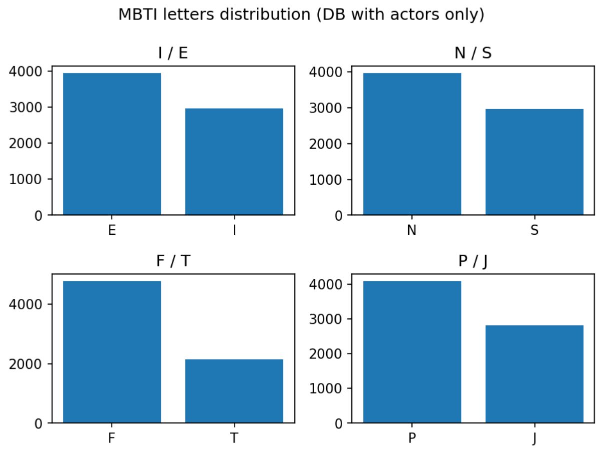

Figure 6: Letter distribution by position (for actors)

Position	Letter A	Letter B	Ratio
1st letter	E: 3 942	I: 2 972	57.0% - 43.0%
2nd letter	N: 3 962	S: 2 952	57.3% - 42.7%
3rd letter	F: 4 765	T: 2 149	68.9% - 31.1%
4th letter	P: 4 090	J: 2 824	59.2% - 40.8%

Table 11: Letter distribution by position (for actors)

Compared to the previous part, we observe a dominance of letter F in 3rd position, indicating actors are on average more Feeling-oriented than the global distribution. For other letters, larger gaps appear without being directly significant. We check letter correlations given the different distribution:

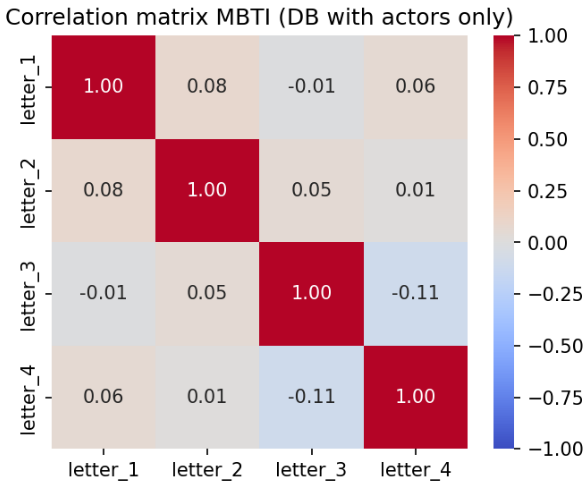

Figure 7: Correlation matrix (for actors)

A slight trend appears between 3rd and 4th letters. We compute p-values and Cramér's V to test significance:

χ^2 Test	p-value	Cramér's V
letter_1 - letter_2	0.0000	0.0789
letter_1 - letter_3	0.4544	0.0090
letter_1 - letter_4	0.0000	0.0550
letter_2 - letter_3	0.0001	0.0480
letter_2 - letter_4	0.4069	0.0100
letter_3 - letter_4	0.0000	0.1076

Table 12: p-value and Cramér's V tests (for actors)

Unlike the entire database part, we have a higher and significant Cramér's V between letters 3 and 4, supporting our correlation matrix. To check parameter dependence, we examine conditional probability:

The conditional distribution between T/F and J/P dimensions highlights a predominance of profile P in both groups. However, this trend is stronger among T individuals (67%) than F individuals (56%). Although the χ^2 test indicates statistical dependence between these variables, Cramér's V remains weak though larger than before, suggesting limited association.

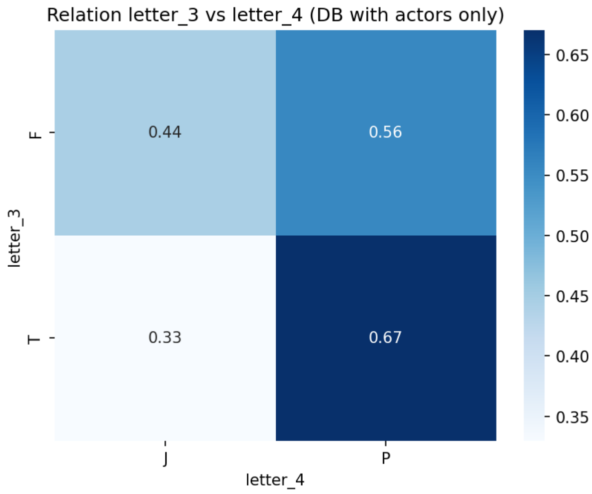

Figure 8: Conditional probability of letters 3 and 4 (for actors)

2.4 Network Construction

To build our network, we iterate over movies within our target years, extracting a chosen number of actors per movie (constant across movies). Using this actor list, we create a graph with edges between actors who played in the same movie. As our graph is unweighted, repeated links across movies are ignored. We vary the extracted actor count to observe evolution between total actors extracted and link count.

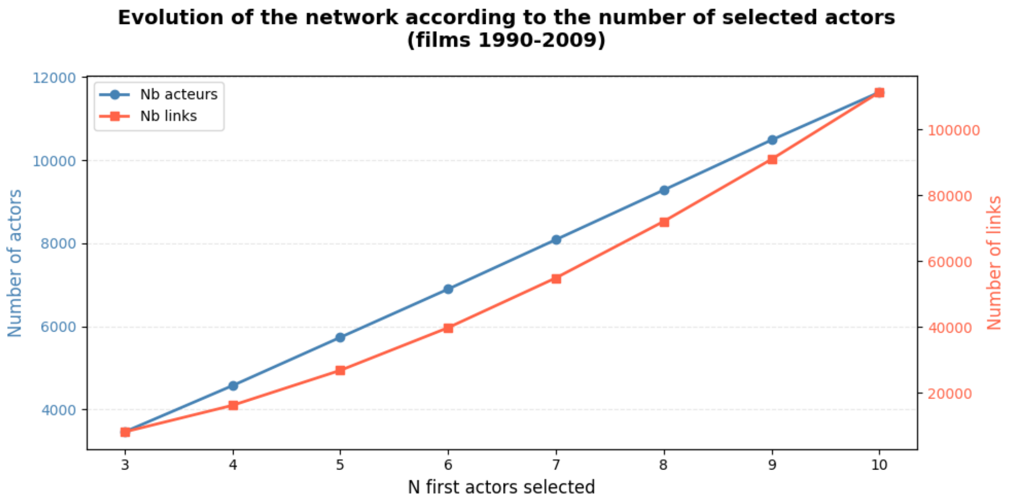

Figure 9: Comparison of actor and link counts vs. extracted actors per movie

Unique actors grow linearly, whereas link counts grow supralinearly.

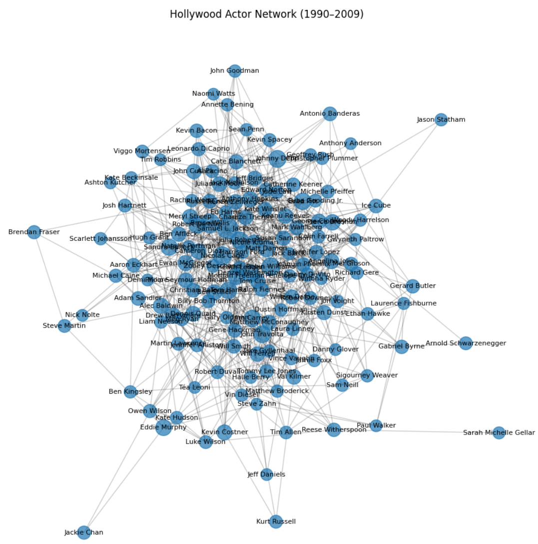

Figure 10: Illustration of constructed network (simplified to nodes with degree ≥ 20)

2.4.1 Actor Centrality

To observe network evolution, we check actor centrality as a function of extracted actors. Here we show mean co-actors (average degree), while `dataset.ipynb` also displays top-3 betweenness and PageRank.

Extracted actors	**Unique actors**	**Number of links**	**Average co-actors**
3	3 455	8 119	4.7
4	4 569	16 184	7.1
5	5 732	26 811	9.4
6	6 894	39 772	11.5
7	8 086	54 894	13.6
8	9 281	72 063	15.5
9	10 486	90 936	17.3
10	11 637	111 299	19.1

Table 13: Evolution of network average degree vs. extracted actors

2.4.2 Small-World Phenomenon

We examine shortest paths between actors and whether a small-world phenomenon is observable:

Extracted actors	Average minimum distance	Clustering coefficient
3	4.62	0.6207
4	4.24	0.6731
5	4.01	0.6993
6	3.85	0.7173
7	3.77	0.7338
8	3.70	0.7450
9	3.61	0.7545
10	3.55	0.7610

Table 14: Evolution of small-world properties vs. extracted actors

A network exhibits small-world properties if average node distance is low and clustering coefficient is high. With clustering $> 60\%$ and average distance < 5, initial conditions already show small-world phenomena. Increasing extracted actors amplifies this phenomenon.

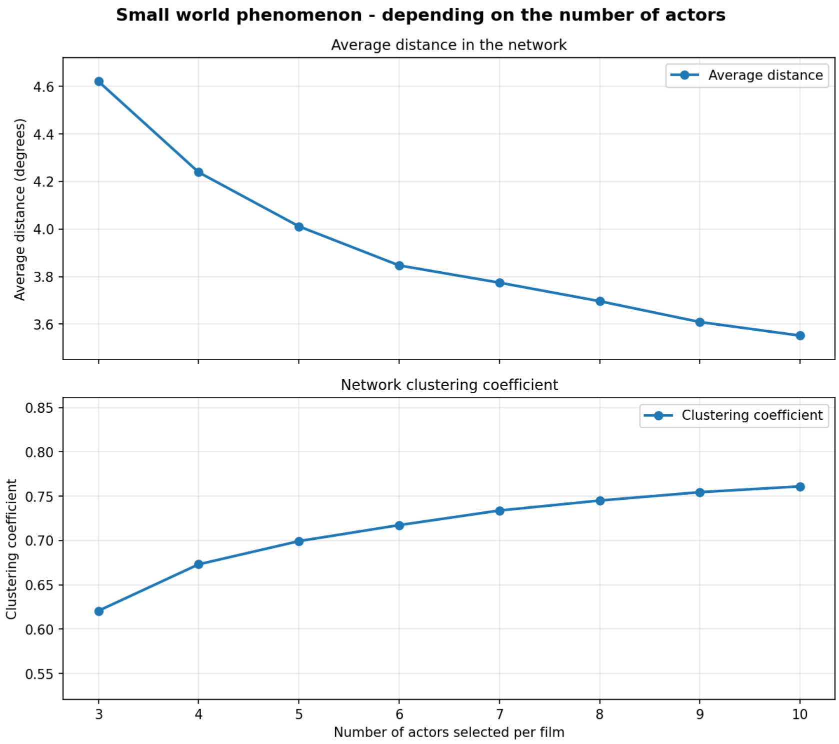

Figure 11: Evolution of small-world phenomena (distance and clustering)

2.4.3 Frequent Actor Duos

We inspect actor pairing frequency. Unlike previous analyses, this requires a weighted graph counting shared movies.

Present binaries originate from multi-film sagas: Harry Potter, Star Trek, X-Men. While interesting, these details are unnecessary here and omitted.

2.4.4 Network Evolution Over Time

Working with movies from 1990 to 2009, global yearly information remains insightful.

Decade	Actors	Links	Density	Main component
1960	182	213	0.0129	17
1970	264	325	0.0094	56
1980	633	807	0.0040	228
1990	1238	2294	0.0030	939
2000	2835	5889	0.0015	1962
2010	2389	4139	0.0015	1425

Table 15: Network evolution by decade

Actor count represents network nodes (actors in ≥ 1 movie that decade) and links represent actor pairs collaborating in a movie. Strong growth reflects database and cinema expansion.

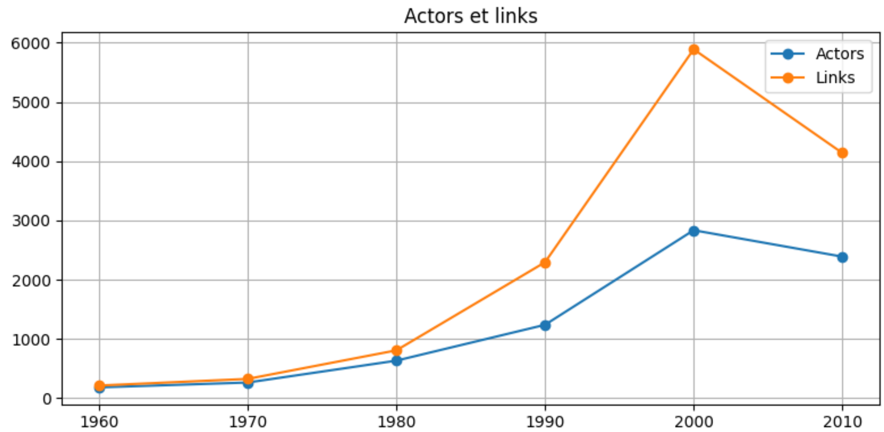

Figure 12: Actor network evolution over decades

Density is existing links over possible links ($\frac{N(N-1)}{2}$ theoretically). Paradoxically, as the network grows, it dilutes because possible collaborations grow faster than actual ones.

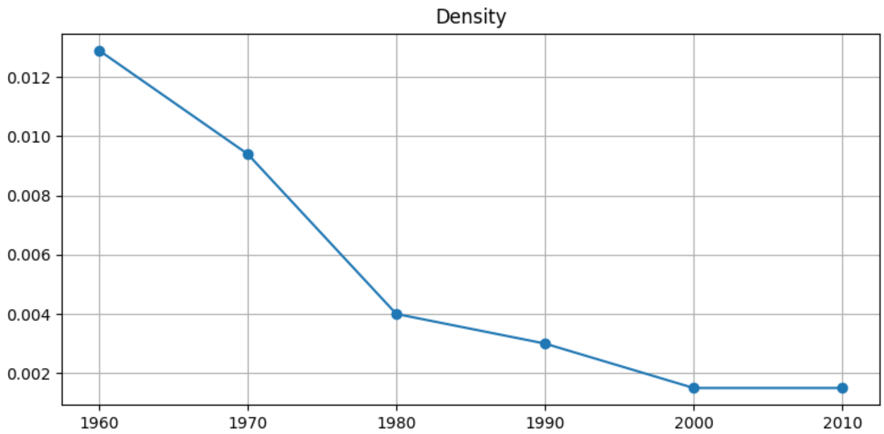

Figure 13: Density evolution over decades

In 1960, only 17 of 182 actors (≈ 9.3%) form the connected core. In 2000, 1962 of 2835 (≈ 69.2%). The network shifts from isolated groups to a large interconnected structure, typical of small-world emergence.

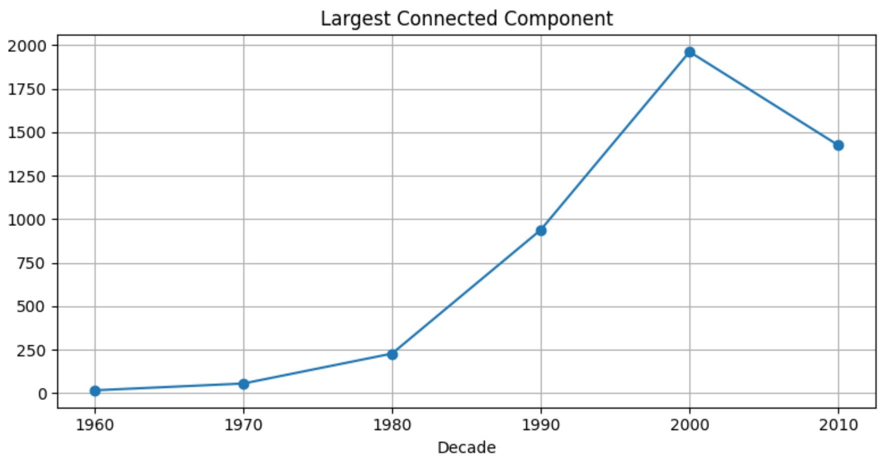

Figure 14: Connected structure evolution over decades

2.5 Final Network

To maximize matching with our MBTI database and obtain better analysis results, we use our network with 10 extracted actors.

2.5.1 Error Correction

Network analysis revealed inconsistencies as "self-loops" (nodes pointing to themselves), numbering 9 for:

- James Caan
- Corey Burton
- David Lascher
- Leo Rossi
- Chad Donella
- Richard Linklater
- Don Austen
- Nick Jameson
- Angelina Idnani

Investigation revealed database entry errors (same character played, ID changed) or actors playing multiple roles in one movie. Both create self-loops upon network construction. We remove these minor links to prevent result distortion.

2.5.2 Network and MBTI Join

We examine network person count and MBTI database length:

Category	**Number**
Network actors	11 637
MBTI celebrities	50 640

Table 16: Network and MBTI database statistics

Name-based matching aligns $2,401$ network individuals (20.6%). Of these, only $1,912$ are defined as actors in our MBTI DB, as several originated elsewhere (chiefly comedians).

To increase personality matches, we use `difflib`'s **get_close_matches** with 90% similarity threshold. Name normalization yielded no better relevance, so we proceeded without it. This produced 223 matches, but inspection revealed spelling variants and distinct individuals.

We performed manual normalization on this subset to prevent erroneous matches, at the cost of omitting some correct ones. We removed punctuation, accents, converted to lowercase, and collapsed whitespace. This matched 45 additional actors, reaching:

Category	**Number**
Network actors	11 637
MBTI celebrities	50 640
Matches	2 446

Table 17: Correspondence between databases

Fuzzy-names

Name matching alternatives existed. Unable to include LLMs directly in code for free, we tested the GitHub *fuzzy-names* repository adapted to Python via LLM.

It operated faster and found relevant matches, but retained false pairings. Filtering for correct answers yielded 42 matches—fewer than our manual process. We retained manual matching while keeping code in `mbti.ipynb`.

2.5.3 Network Distribution

With matched personalities in our network, we analyze the MBTI distribution:

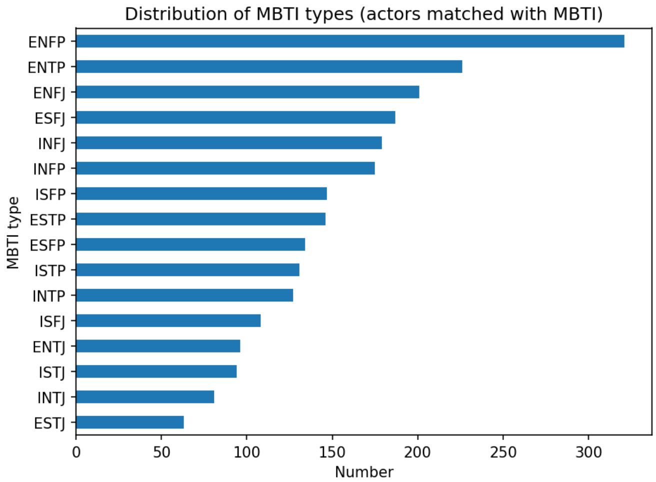

Figure 15: MBTI type distribution (network match)

MBTI Type	Count	Percentage
ENFP	321	13.3%
ENTP	226	9.4%
ENFJ	201	8.3%
ESFJ	187	7.7%
INFJ	179	7.4%
INFP	175	7.2%
ISFP	147	6.1%
ESTP	146	6.0%
ESFP	134	5.5%
ISTP	131	5.4%
INTP	127	5.3%
ISFJ	108	4.5%
ENTJ	96	4.0%
ISTJ	94	3.9%
INTJ	81	3.4%
ESTJ	63	2.6%

Table 18: MBTI type distribution (network match)

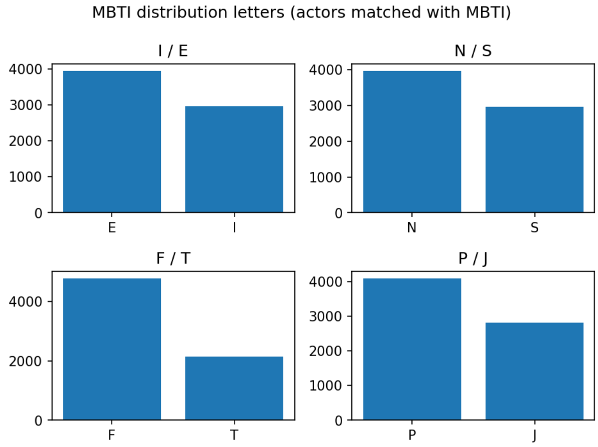

Figure 16: Letter distribution by position (network match)

Position	Letter A	Letter B	Ratio
1st letter	E: 1 374	I: 1 042	56.9% - 43.1%
2nd letter	N: 1 406	S: 1 010	58.2% - 41.8%
3rd letter	F: 1 452	T: 964	60.1% - 39.9%
4th letter	P: 1 407	J: 1 009	58.2% - 41.8%

Table 19: Letter distribution by position (network match)

As with actors, letter `F` dominates in 3rd position. Our match network reflects observed actor trends. Correlation analyses ensure no dependency was introduced:

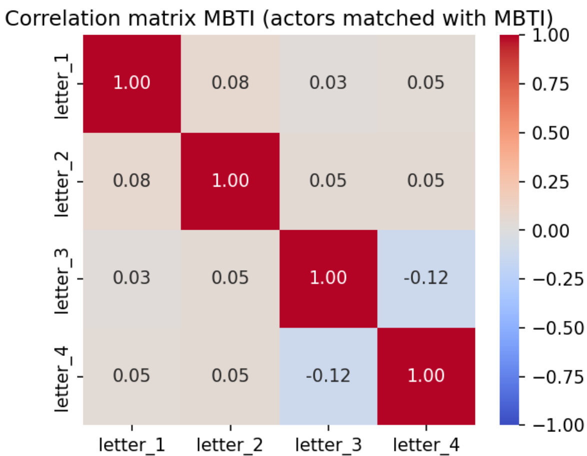

Figure 17: Correlation matrix (network match)

A slightly more significant value appears between letters 3 and 4, leading us to compute p-values and Cramér's V.

χ^2 Test	p-value	Cramér's V
letter_1 - letter_2	0.0003	0.0744
letter_1 - letter_3	0.1604	0.0286
letter_1 - letter_4	0.0283	0.0446
letter_2 - letter_3	0.0102	0.0523
letter_2 - letter_4	0.0130	0.0505
letter_3 - letter_4	0.0000	0.1167

Table 20: p-value and Cramér's V tests (network match)

Consistent with actor data, Cramér's V is slightly more significant between letters 3 and 4. We observe conditional probability to check potential links:

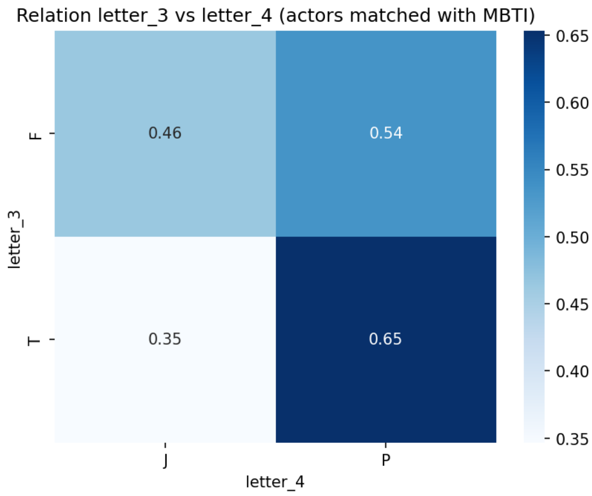

Figure 18: Conditional probability of letters 3 and 4 (network match)

Although the χ^2 test indicates statistical dependence, Cramér's V remains weak, suggesting limited association. Dependency cannot be asserted as concluded for actor data.

2.6 Evaluation

2.6.1 Independence Calculation

χ^2 *Test*

The χ^2 test of independence determines if a relationship exists between two dimensions, verifying whether variable categories are linked or independent.

Hypotheses formulation:

- Null hypothesis (H_0): Variables are independent (no relationship).
- Alternative hypothesis (H_1): Variables are not independent.

Given N samples and observed frequency O_{ij} where $X = i$ and $Y = j$, expected value E_{ij} under independence is:

$$E_{ij} = \frac{O_{i+} \times O_{+j}}{N}$$

Where:

$$O_{i+} = \sum_{j=1}^{J} O_{ij}$$

(data where $X = i$)

$$O_{+j} = \sum_{i=1}^{I} O_{ij}$$

(data where $Y = j$)

The χ^2 statistic is:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Low χ^2 suggests independence; high χ^2 indicates potential association.

P-value

The p-value is the probability of obtaining a χ^2 at least as high as observed under H_0. Low p-value (≤ 0.05) indicates observed deviations are unlikely due to chance.

However, large datasets can yield p-value ≈ 0 even for weak relations, necessitating Cramér's V.

Cramér's V

Cramér's V evaluates the strength of association between qualitative variables from 0 to 1. Values near 0 (≤ 0.1) indicate weak/no association; values near 1 indicate strong relations.

2.6.2 Homophily

Homophily is the tendency to bond with similar individuals ("similarity breeds connection"). We test this on our network.

2.6.2.1 Transition Matrices

These quantify connection probabilities between types. For each dimension pair ($src\_dim \longrightarrow tgt\_dim$), we build a matrix answering:

"Given node u has $from\_val$ on source dimension, what is probability neighbor v has to_val on target dimension?"

With 4 dimensions (letters), we obtain 16 matrices (2×2 or 3×3 depending on "Unknown").

For each edge $(u, v) \in E$, we count co-occurrences:

$$C(src\_dim, tgt\_dim, from\_val, to\_val) = \sum_{(u,v) \in E} 1[u_{src} = from\_val] \cdot 1[v_{tgt} = to\_val]$$

Total outgoing links:

$$C_{total}(src\_dim, tgt\_dim, from\_val) = \sum_{to\_val} C(src\_dim, tgt\_dim, from\_val, to\_val)$$

Conditional probability:

$$P(to\_val_c|from\_val_r) = \frac{C(src\_dim, tgt\_dim, from\_val_r, to\_val_c)}{C_{total}(src\_dim, tgt\_dim, from\_val_r)}$$

Lift

To evaluate over/under-representation, we compute marginal distribution and lift:

$$P_{marg}(to\_val_c) = \frac{\sum_r C(src\_dim, tgt\_dim, from\_val_r, to\_val_c)}{\sum_r C_{total}(src\_dim, tgt\_dim, from\_val_r)}$$

$$Lift(from\_val_r) = \frac{1}{|C|}\sum_c \frac{P(to\_val_c|from\_val_r)}{P_{marg}(to\_val_c)}$$

Lift	**Significance**
$> 1.0\times$	Nodes $from\_val$ connect more than expected by chance
$= 1.0\times$	Distribution conforms to chance
$< 1.0\times$	Nodes $from\_val$ connect less than expected by chance

Table 21: Lift coefficient interpretation

2.6.2.2 Degree Centrality

Node degree measures direct connections:

$$k_u = \sum_{v\in V} 1[(u, v) \in E]$$

Mean Degree

$$k_{moy}(dim, val) = \frac{1}{|\mathcal{N}|}\sum_{u\in\mathcal{N}} k_u$$

Median Degree

$$k_{med}(dim, val) = \text{median}(\{k_u|u \in \mathcal{N}\})$$

2.6.2.3 Betweenness Centrality

Measures how often node u falls on shortest paths:

$$C_B(u) = \sum_{s\neq u\neq t\in V} \frac{\sigma(s, t \mid u)}{\sigma(s, t)}$$

Normalized:

$$C_B^{norm}(u) = \frac{C_B(u)}{(|V| - 1)(|V| - 2)/2}$$

2.6.2.4 Closeness Centrality

$$C_C(u) = \frac{1}{d(u)} = \frac{|V| - 1}{\sum_{v \neq u \in V} d(u, v)}$$

2.6.2.5 PageRank Centrality

$$PR(u) = \frac{1 - \alpha}{|V|} + \alpha \sum_{v \in \mathcal{N}(u)} \frac{PR(v)}{k_v}$$

with damping factor $\alpha = 0.85$.

2.6.2.6 Eigenvector Centrality

$$C_E(u) = \frac{1}{\lambda} \sum_{v \in \mathcal{N}(u)} C_E(v)$$

2.6.2.7 Clustering Coefficient

$$C_{clust}(u) = \frac{|\{(v, w) \in E \mid v, w \in \mathcal{N}(u)\}|}{k_u(k_u - 1)/2}$$

Global average:

$$\bar{C}_{clust} = \frac{1}{|V|} \sum_{u \in V} C_{clust}(u)$$

2.6.2.8 Rich Club Coefficient

$$\varphi(k) = \frac{2 \cdot |E_{>k}|}{|\mathcal{V}_{>k}| \cdot (|\mathcal{V}_{>k}| - 1)}$$

2.6.3 Heterophily

Heterophily measures tendency to bond with different profiles.

2.6.3.1 Heterophily Score

For each movie, top $n = 10$ actors are intersected with MBTI DB. Movies with < 2 matched actors are excluded. Shannon entropy of binary variables measures diversity:

$$H(p) = -\left[p \log_2(p) + (1 - p) \log_2(1 - p)\right]$$

Score

$$\text{hetero\_score(film)} = \frac{1}{4}\Big[H(p_E) + H(p_S) + H(p_T) + H(p_J)\Big] \in [0, 1]$$

$$\text{Category(film)} = \begin{cases} \text{Homogeneous} & \text{hetero\_score} < 0.25 \\ \text{Mixed} & 0.25 \leq \text{hetero\_score} \leq 0.75 \\ \text{Heterogeneous} & \text{hetero\_score} > 0.75 \end{cases}$$

2.6.3.2 Community Formation

To isolate specificity independent of population bias, centered deviations are computed:

$$\delta_E(\text{film}) = p_E(\text{film}) - \bar{p}_E$$

K-Means

K-Means partitions the n films into k clusters by minimizing intra-cluster inertia:

$$J(k) = \sum_{i=1}^{k} \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2$$

where $\boldsymbol{\mu}_i$ is the centroid of the cluster C_i:

$$\boldsymbol{\mu}_i = \frac{1}{|C_i|} \sum_{\mathbf{x} \in C_i} \mathbf{x}$$

The algorithm iteratively alternates between assignment (each point to the nearest centroid) and update (recalculating centroids) until convergence.

This method is performed by varying the number of clusters, and then the elbow method is used to select the optimal number of clusters. The inertia $J(k)$ decreases monotonically with k (more clusters always reduce the inertia). The network library kneed is then used to detect the point of maximum curvature on the curve $(k, J(k))$.

DBSCAN

DBSCAN groups points based on local density: a point belongs to a cluster if it has enough close neighbors. Points that are too isolated are labeled as noise (-1).

Our data comes from proportions on small samples. We produce a continuous and relatively uniform point cloud in space, without any real "density pockets" separated by empty areas. We can see that for our partitioning, we obtain, in descending order:

Nom du cluster	Nombre de films
Groupe 0	2263
Groupe -1 (bruit)	121
Groupe 12	24
Groupe 15	21
Groupe 19	21

Table 22: Répartitions des films par cluster avec DBSCAN

We clearly see from our results that this method is not suitable for our situation, and we will therefore abandon it.

Louvain

Louvain detects communities by maximizing modularity on a graph: it groups nodes that have more internal connections than would be expected by chance. The graph is constructed by connecting films whose Euclidean distance is below a certain threshold.

This method is designed for networks where connections are primary data. In our case, the graph is entirely derived from a Euclidean distance in only 4 dimensions. By manipulating the parameters of our method, we obtained a partitioning into 29 communities, but these were often of similar size. These results indicate a rather random and irrelevant grouping, which is why we continued using only the K-means method.

3 Results

3.1 Homophily

We analyze network links among personality types, distinguishing cases with and without unknown personality data.

3.1.1 Complete Network

3.1.1.1 Transition

Starting from 3 values per dimension plus "Unknown", we obtain sixteen 3×3 transition matrices with lift.

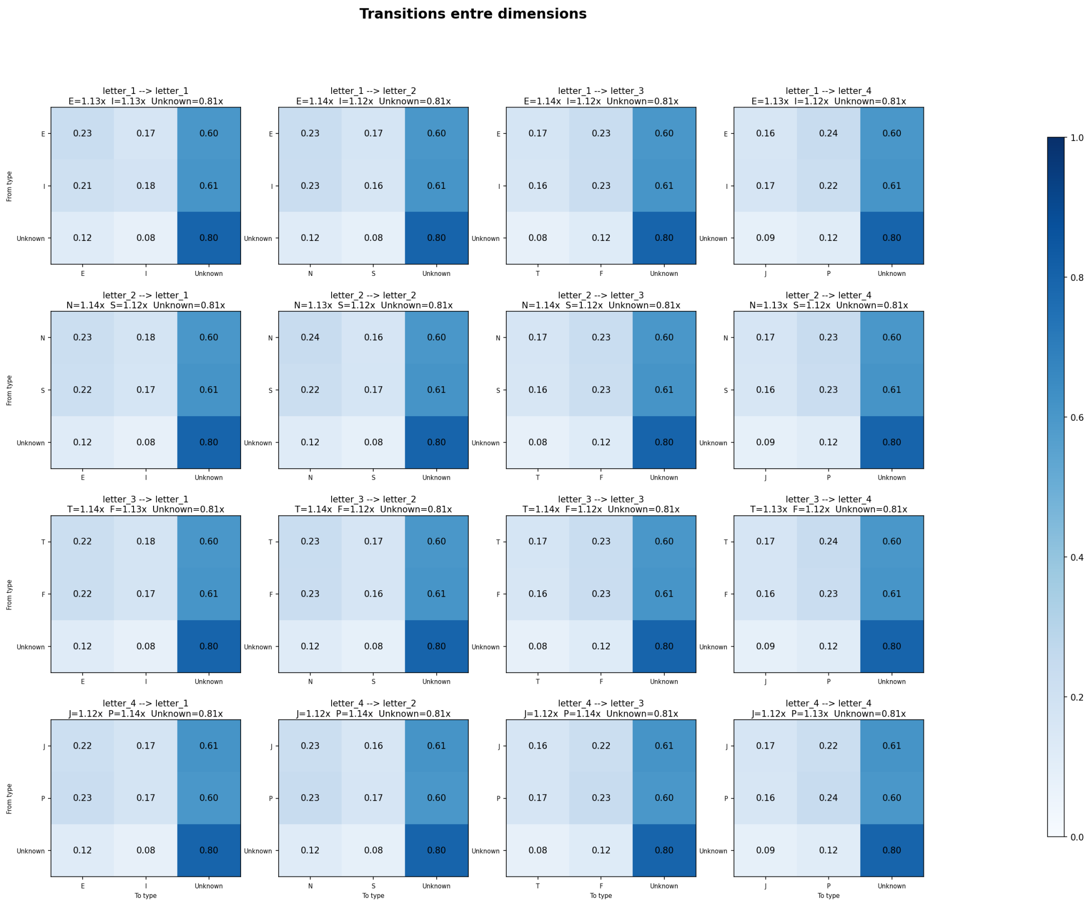

Figure 19: Transition matrices (full network)

"Unknown" dominates due to unmatched nodes. Defined letters show higher lift, reflecting more prominent, well-known recurring actors.

3.1.1.2 Degree Centrality

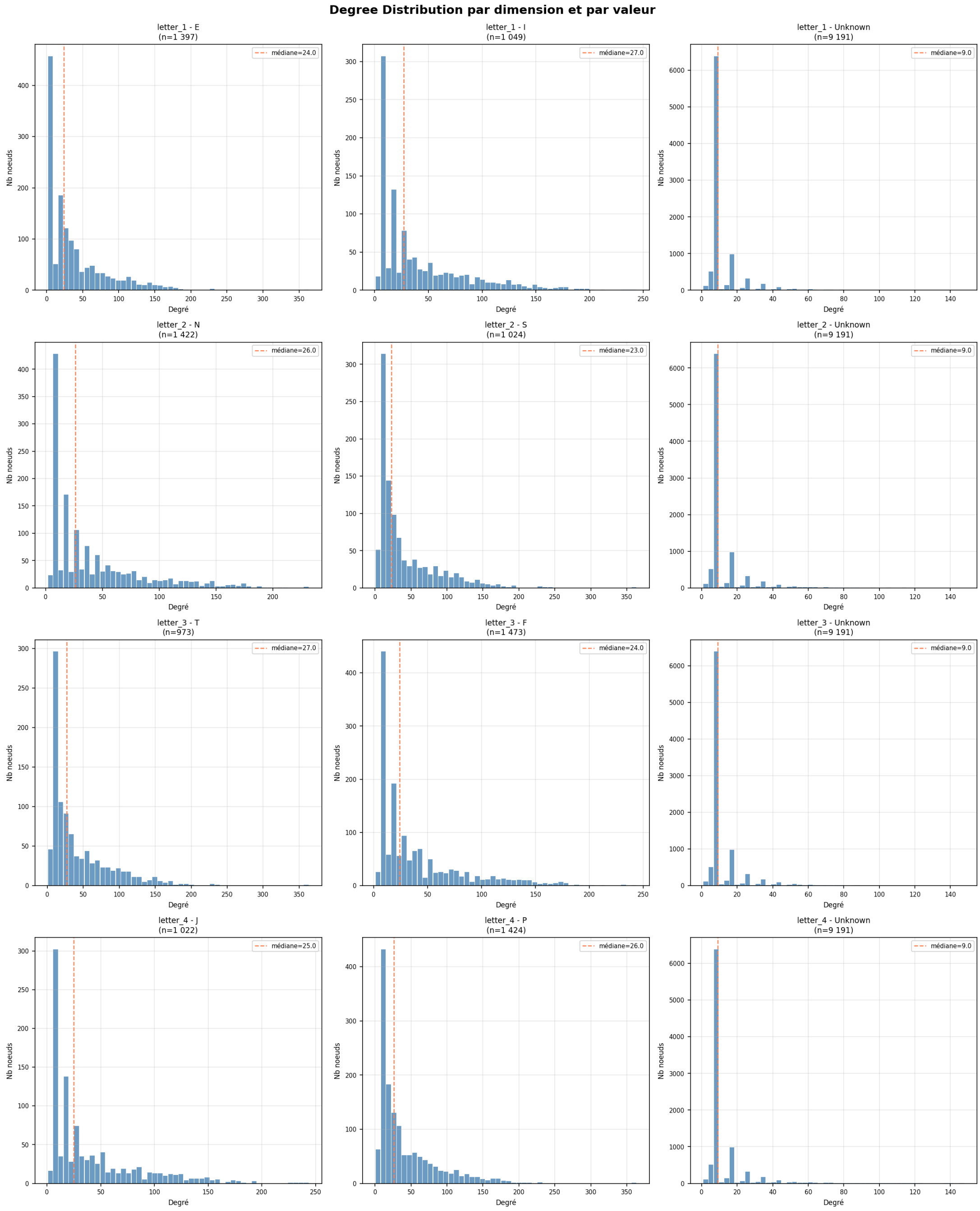

Figure 20: Degree distribution (full network)

As expected, all our curves are downward sloping, with many nodes containing few links and fewer nodes as we increase the degree. The major difference lies in the graphs of the "Unknown" value, where, although much more numerous (9191 nodes versus $\sim$ 2500), we obtain a median $\sim$ 3x smaller than that of the defined nodes. We can therefore conclude that the undefined nodes are actors with fewer collaborations and thus certainly less well-known and less popular actors.

3.1.1.3 Betweenness Centrality

The betweenness centrality of a node measures how close it is to other nodes in a network. If its value is 0, the node is not on any path between other nodes. As expected in most real-world networks, most nodes are "ordinary" and do not represent a central link.

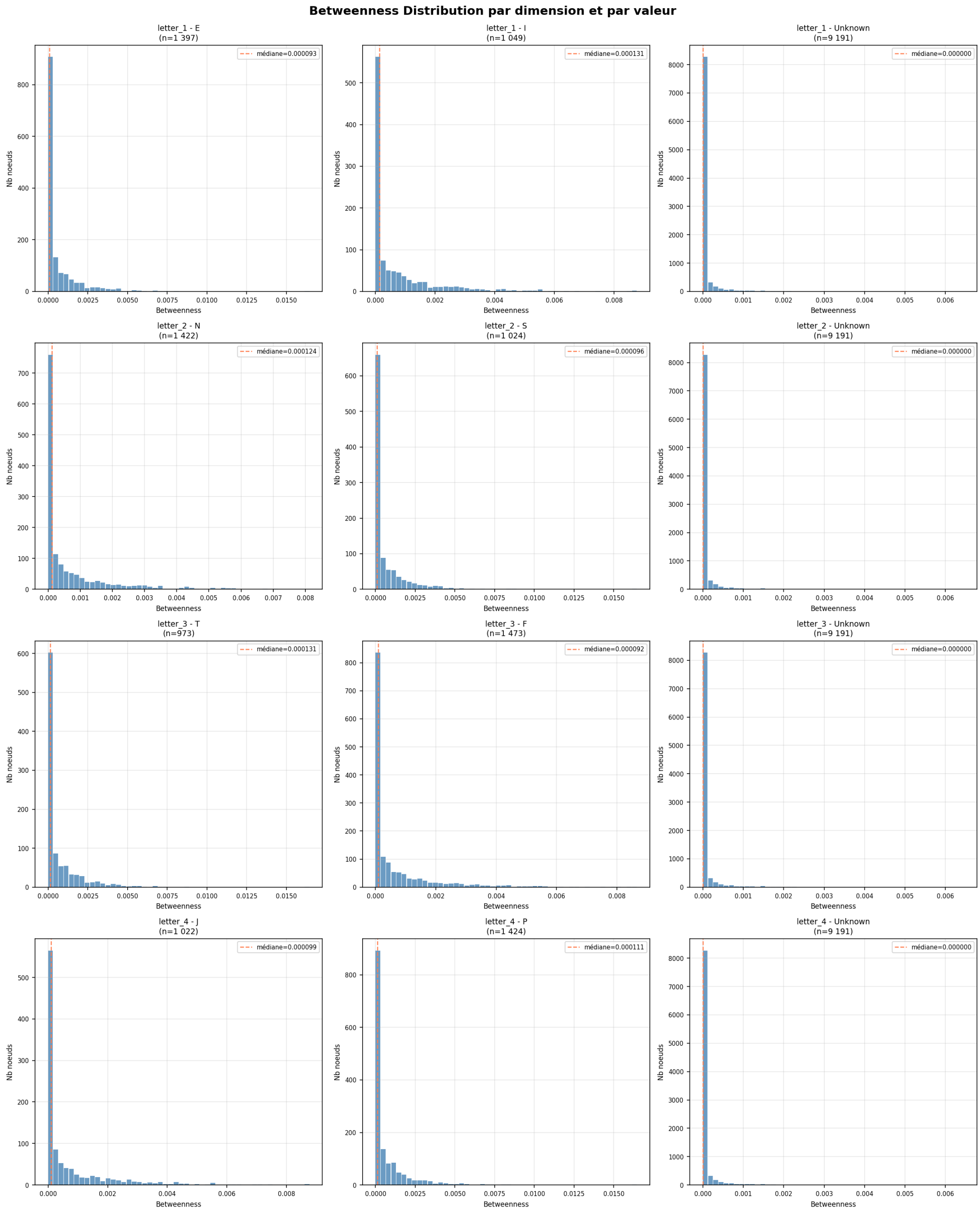

Figure 21: Betweenness distribution (full network)

We note that the median for the letters "Unknown" is 0, supporting the hypothesis that the nodes whose values are unknown are located on the periphery of our network. The rather low value of the betweenness also corresponds to a natural network with only a few central points.

3.1.1.4 Closeness Centrality

Closeness centrality measures how close a node is to all other nodes in the network. Closeness typically follows a normal distribution, meaning that nodes are relatively homogeneous.

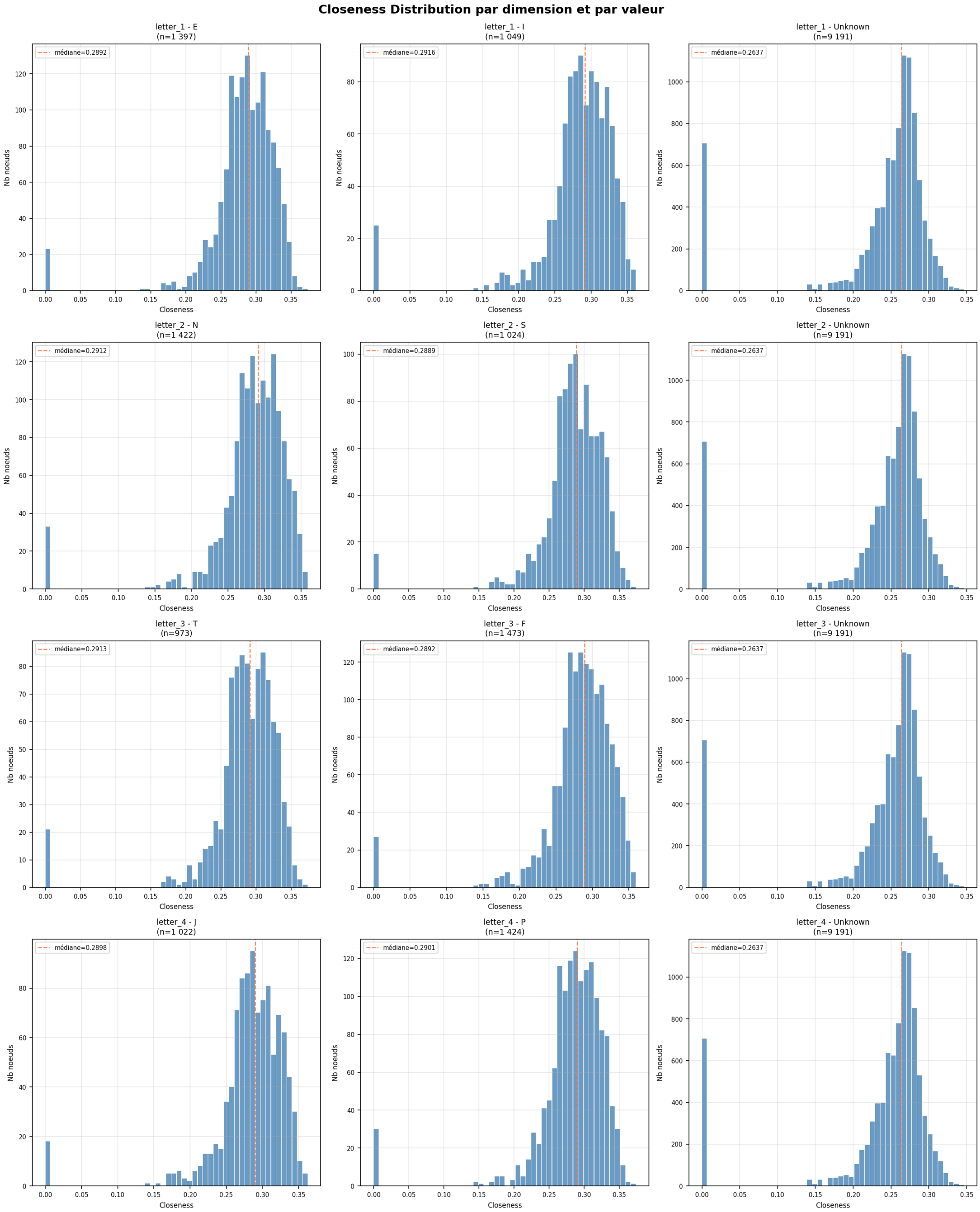

Figure 22: Closeness distribution (full network)

In our graphs, we observe a peak at 0 representing isolated nodes (disconnected components of the graph). This peak is much higher in the "Unknown" graphs, given that these nodes are located at the periphery of the network and are therefore more often isolated. The rest of our curve clearly illustrates a normal distribution around the median, located around ~ 0.29 (~ 0.26 for "Unknown"). These differences are minimal and suggest that the MBTI type has little influence on structural position within the network.

3.1.1.5 PageRank Centrality

In line with our previous results, our PageRank graphs show a high initial peak followed by a gradual decline with a long tail. This is the signature of a network where a few nodes accumulate a lot of influence.

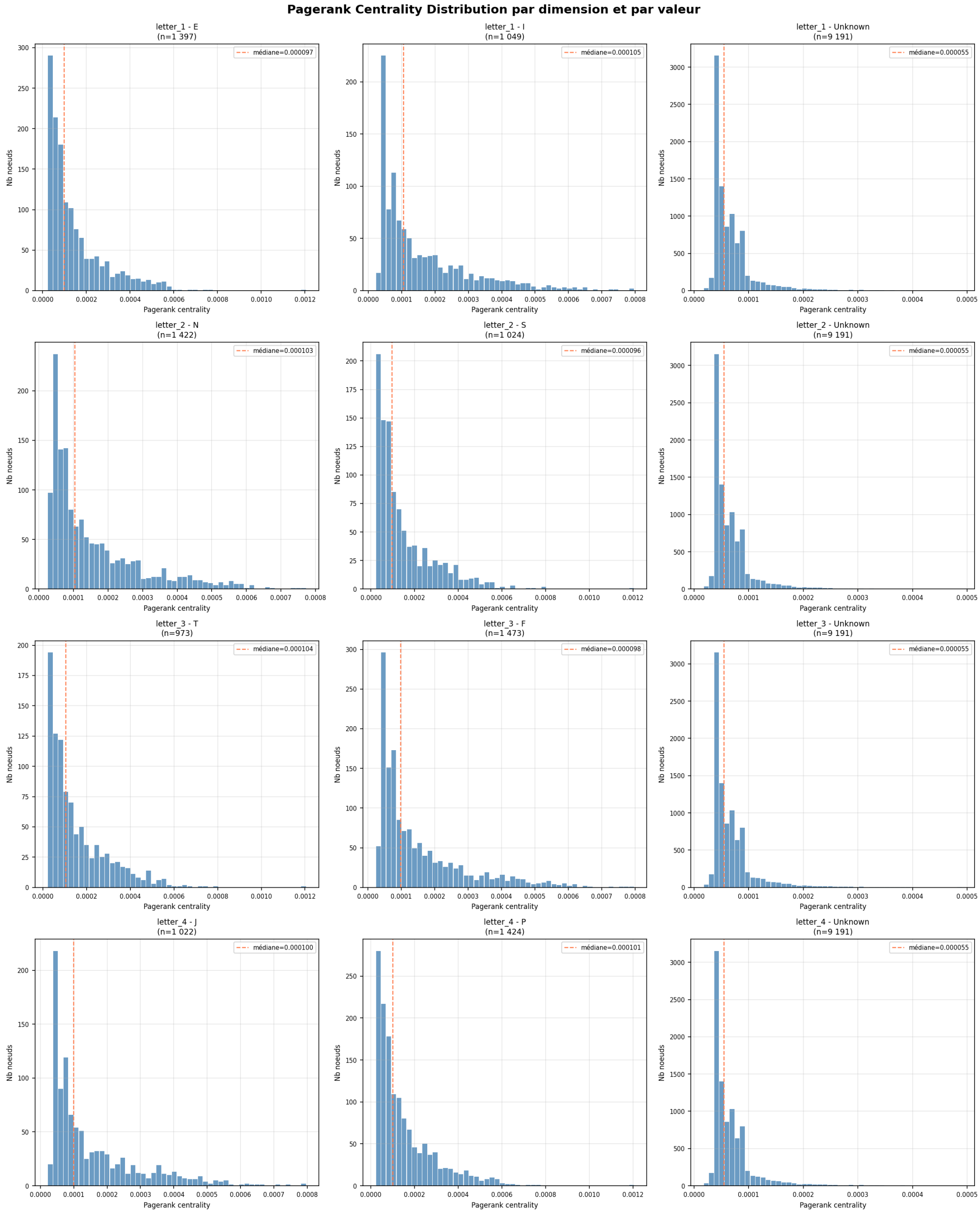

Figure 23: PageRank distribution (full network)

As with proximity, their median is very close (~ 0.0001) and indicates that the MBTI type does not structurally influence the median PageRank. The median of the "Unknown" graphs has a value half that of the median, clearly showing that these nodes are located on the periphery for all metrics.

We note that the letters `E, S, T, P` have a distribution extending up to 0.0012. This means that within these groups, a few nodes have a disproportionate influence well beyond the median, corresponding to the profile of Samuel L. Jackson (the most central node in our network).

3.1.1.6 Eigenvector Centrality

The centrality of the eigenvector measures the influence of a node by taking into account the influence of its neighbors, therefore a node in a dense group of influential nodes explodes in value.

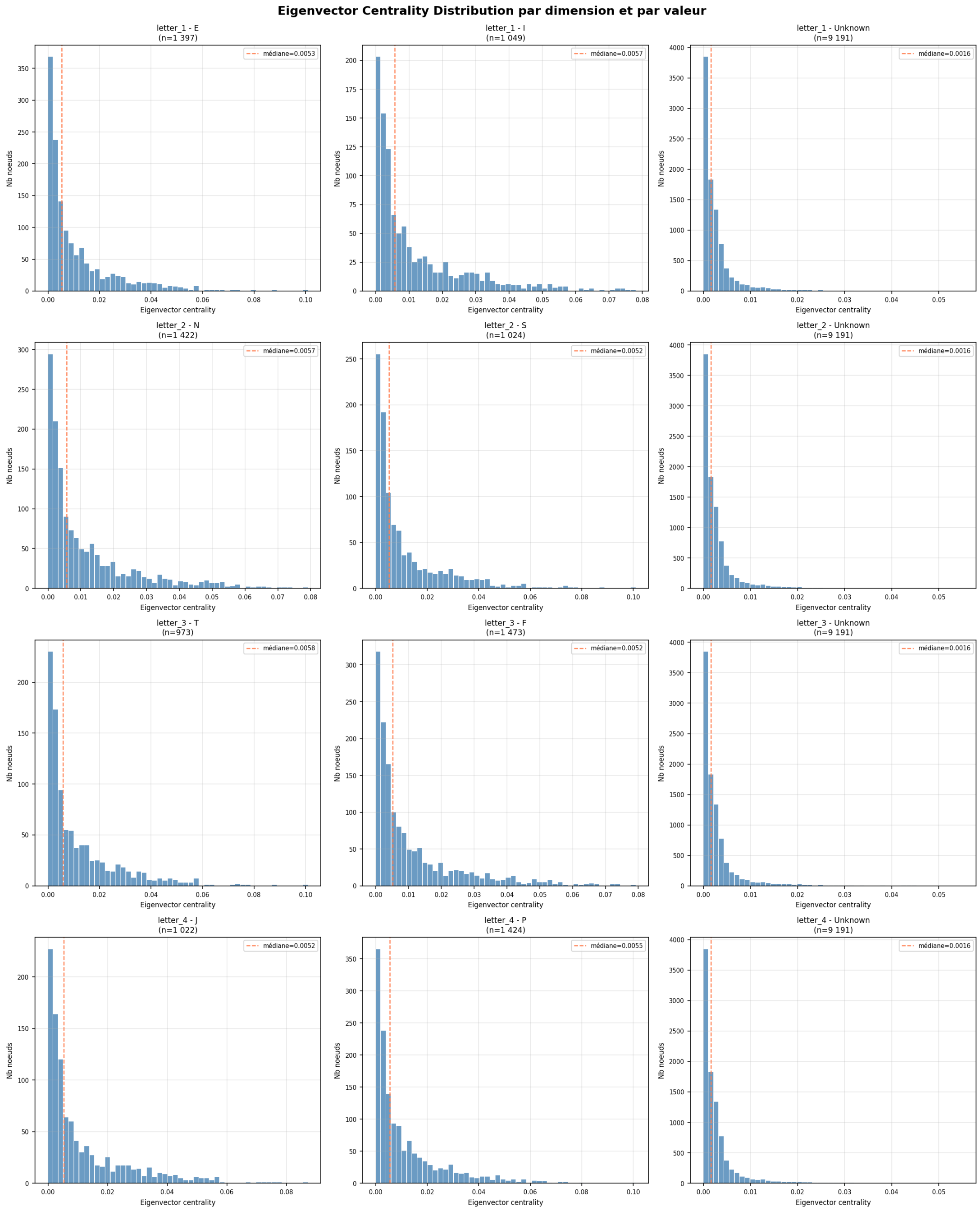

Figure 24: Eigenvector distribution (full network)

As with our other measures, the median is similar for defined letters (~ 0.005) and lower for "Unknown". We can note a slightly higher median for the letters: `I`, `N`, `T`, which suggests that their connections may be more "qualitative," however, the difference is minimal.

3.1.1.7 Clustering Coefficient

Here, we look at how interconnected a node's neighbors are. For better interpretation, we will relate two metrics simultaneously. We are no longer looking at an isolated distribution but at a structural law of the network.

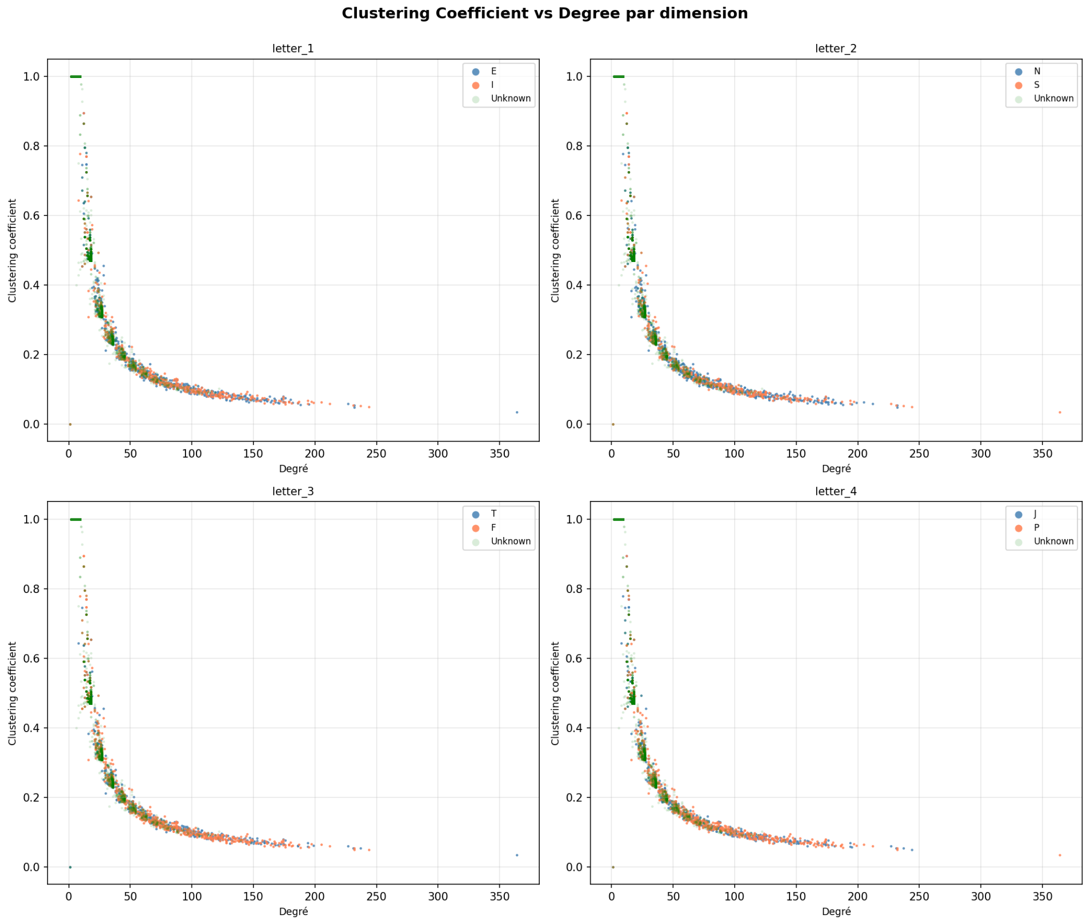

Figure 25: Clustering vs. degree (full network)

We observe a hyperbolic curve, the signature of a scale-free network. We note a plateau with a clustering of 1 for degrees < 10, with the exception of a few points at (0, 0) which are isolated points in the network. We clearly see a predominance of "Unknown" points in degrees < 100. The different letters perfectly overlap on the same curve in all four graphs, thus no MBTI type stands out.

Between degrees 10 and 30, we have a high variability in clustering (0.3 to 0.9). Nodes with similar degrees but very different roles: some in closed cliques, others already bridging communities.

3.1.1.8 Rich Club Coefficient

The Rich Club measures whether highly connected nodes tend to connect with each other rather than with less connected nodes.

Our curve rises steadily from 0 to 1, a hallmark of a progressive and robust Rich Club effect. We observe a sharp increase between degrees 228 and 240: ϕ jumps from 0.4 to 1.0. This indicates the existence of a small group of super-hubs that form a near-perfect clique. The oscillations between 180 and 200 degrees suggest that few nodes are present at these high degrees, and consequently, each node added to or removed from the club has a noticeable impact on $\phi(k)$.

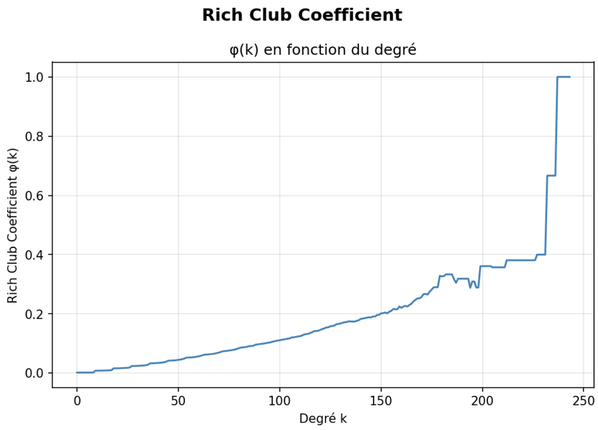

Figure 26: Rich Club vs. degree (full network)

Continuous rise indicates robust rich-club effect, with a sharp jump between degrees 228 and 240 (ϕ from 0.4 to 1.0).

3.1.2 Filtered Network (Without Unknowns)

Excluding unknown MBTI nodes:

	Before filtering		After filtering	
Metric	**Count**	**%**	**Count**	**%**
Nodes	11 637	100.0%	2 446	21.0%
Links	111 290	100.0%	26 126	23.5%

Table 23: Graph comparison before and after filtering

3.1.2.1 Transition

Since we no longer have unknown nodes in our new network, we can represent the transition matrices in size 2×2 to spot potential links.

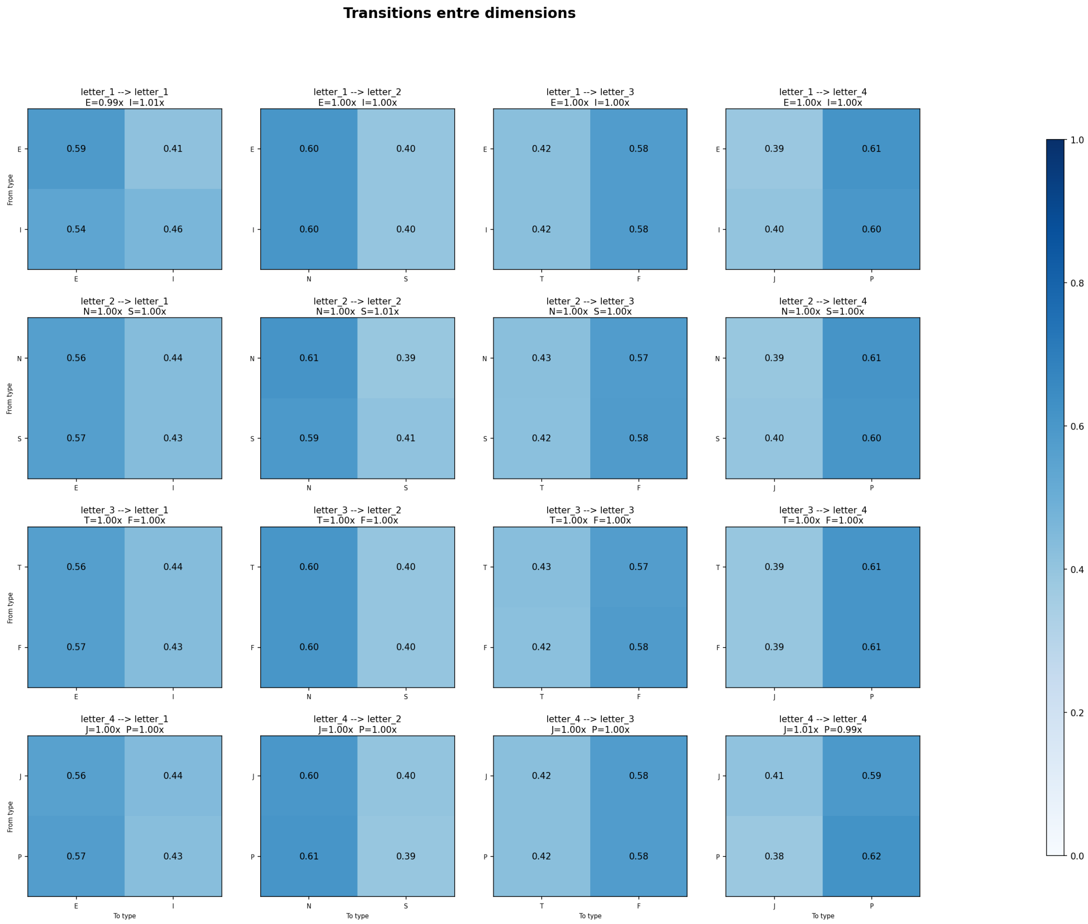

Figure 27: Transition matrices (filtered network)

For our 16 transition matrices, we immediately notice that the lift is always 1 for all our letters, meaning that the observed transition probability is exactly equal to the probability expected by chance. We therefore have a clear result that indicates no link and thus independence between the transitions and the MBTI dimensions.

3.1.2.2 Degree Centrality

Looking at the degree distribution with our filtered network, we can expect to have a power law typical of collaboration networks as with our overall results, with a large majority of actors having a low degree and a minority of actors having very connected

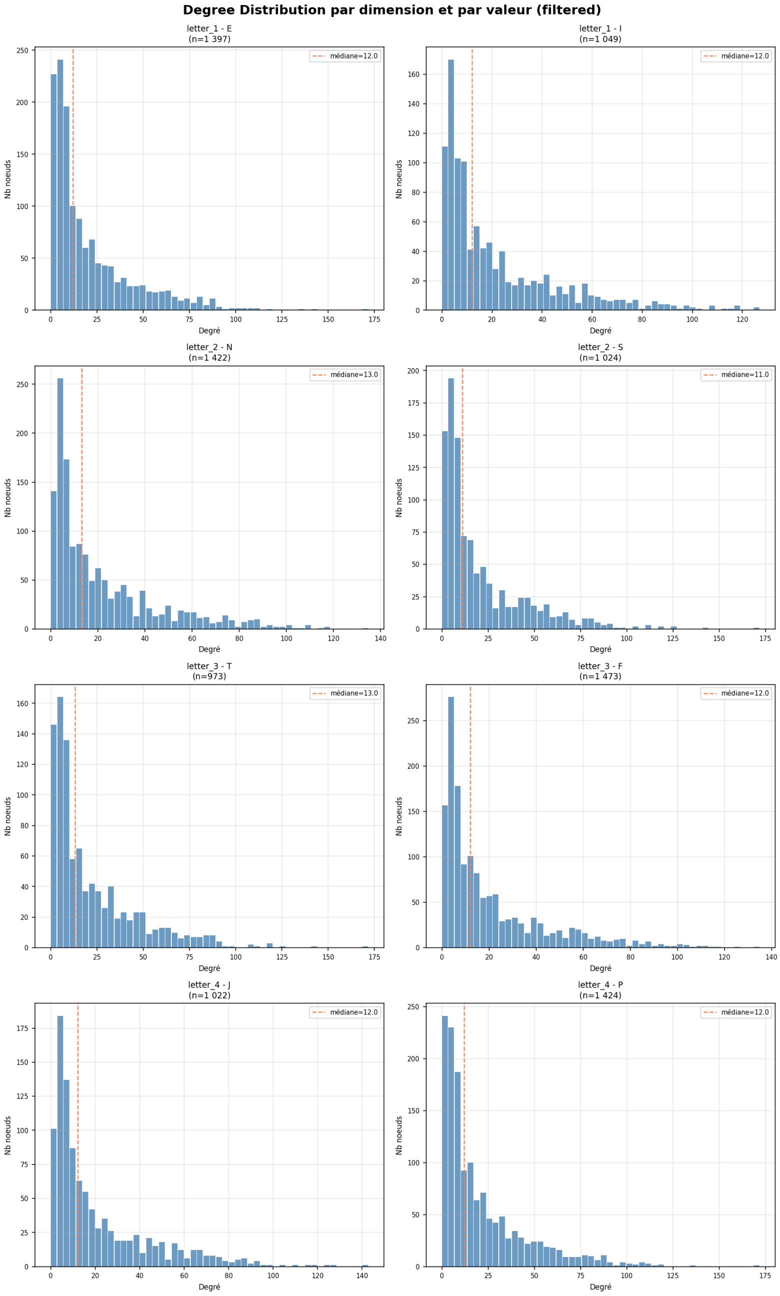

Figure 28: Degree distribution (filtered network)

We note that the median of all our letters is around 12 degrees. Since this value is shared by all our dimensions, we can therefore say that the degree of our nodes is not influenced by the MBTI data. In other words, no abnormal signs appear, and therefore no correlations exist with the degree of our nodes.

3.1.2.3 Betweenness Centrality

As with our results on the full network, we expect a distribution of intermediary centrality with a concentrated peak near zero followed by a very broad, straight tail (power law). This morphology is consistent with the scale-free structure identified during the degree analysis.

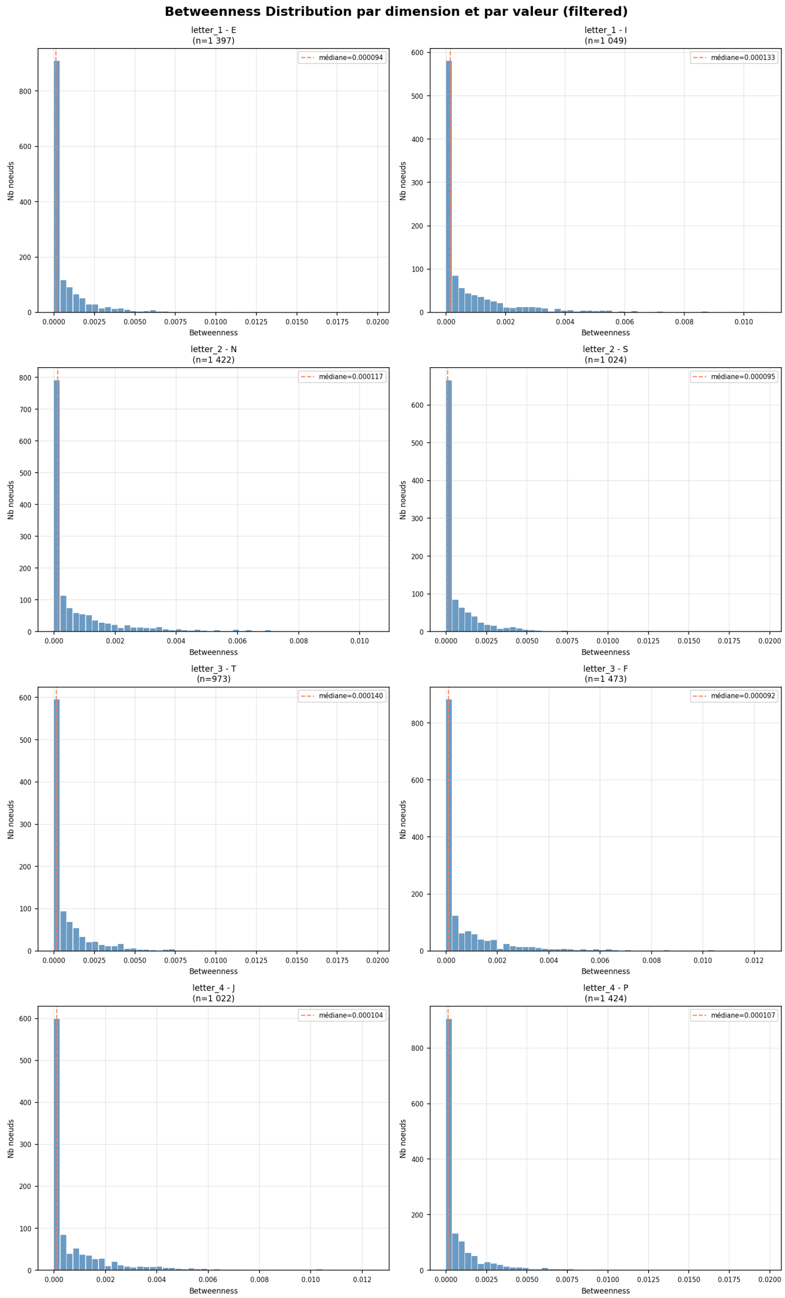

Figure 29: Betweenness distribution (filtered network)

The medians, around 0.0001, are extremely low and uniform across all types. This homogeneity confirms that an actor's MBTI profile is not associated with a particular structural position within the network of collaborations.

Some actors achieve intermediate values of around 0.01 to 0.02, or one hundred to two hundred times the median. Their distribution across MBTI types does not reveal any significant over-representation of a particular profile, which reinforces the hypothesis of independence between personality and structural role.

3.1.2.4 Closeness Centrality

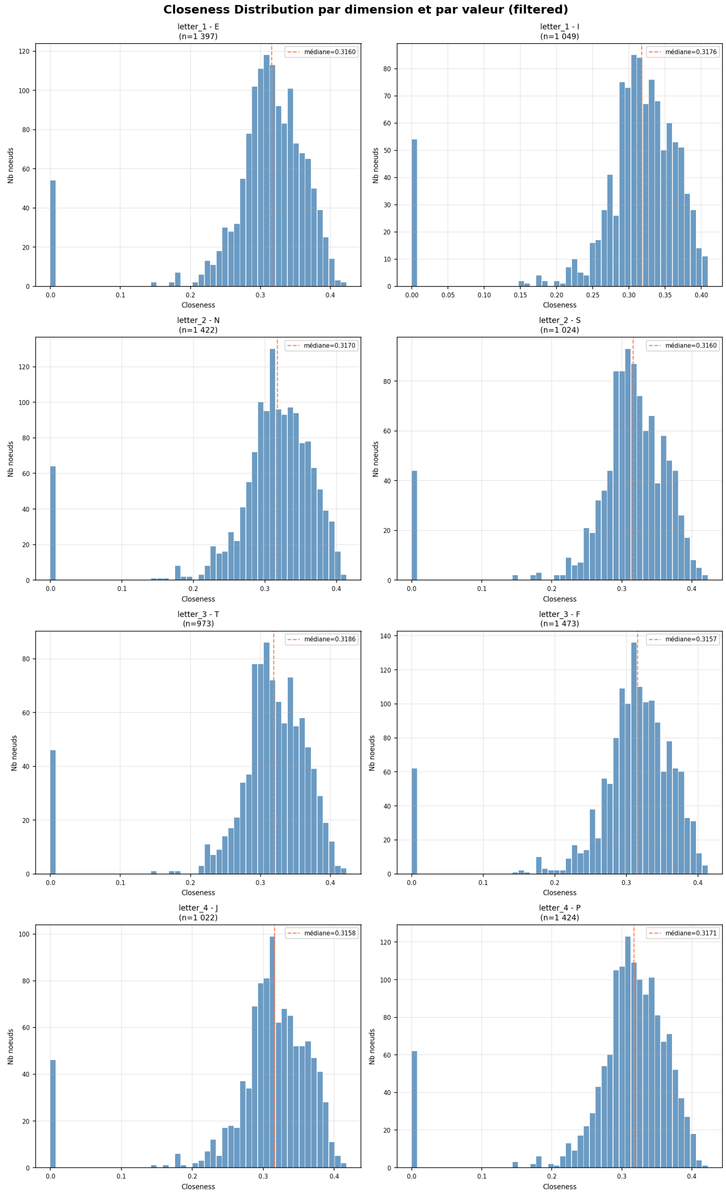

Figure 30: Closeness distribution (filtered network)

All distributions exhibit a relatively symmetrical bell shape, centered around the median of 0.31–0.32, without a pronounced heavy tail. This Gaussian morphology indicates that the actors in the filtered network are generally equidistant from one another: there are no actors structurally much closer to the rest of the network than the average.

As with degree and middleness, all medians fall within an extremely narrow range.

The network thus exhibits a small-world property, characteristic of professional collaboration networks. A secondary peak visible around 0 is common to all distributions. It corresponds to actors located in isolated or weakly connected components. This peak, being more significant than the one present with the complete network, is explained by the significant reduction of the graph during filtering, which may have fragmented some peripheral connections. The proximity centrality provides further confirmation of all the previous results: an actor's proximity is independent of their MBTI profile.

3.1.2.5 PageRank Centrality

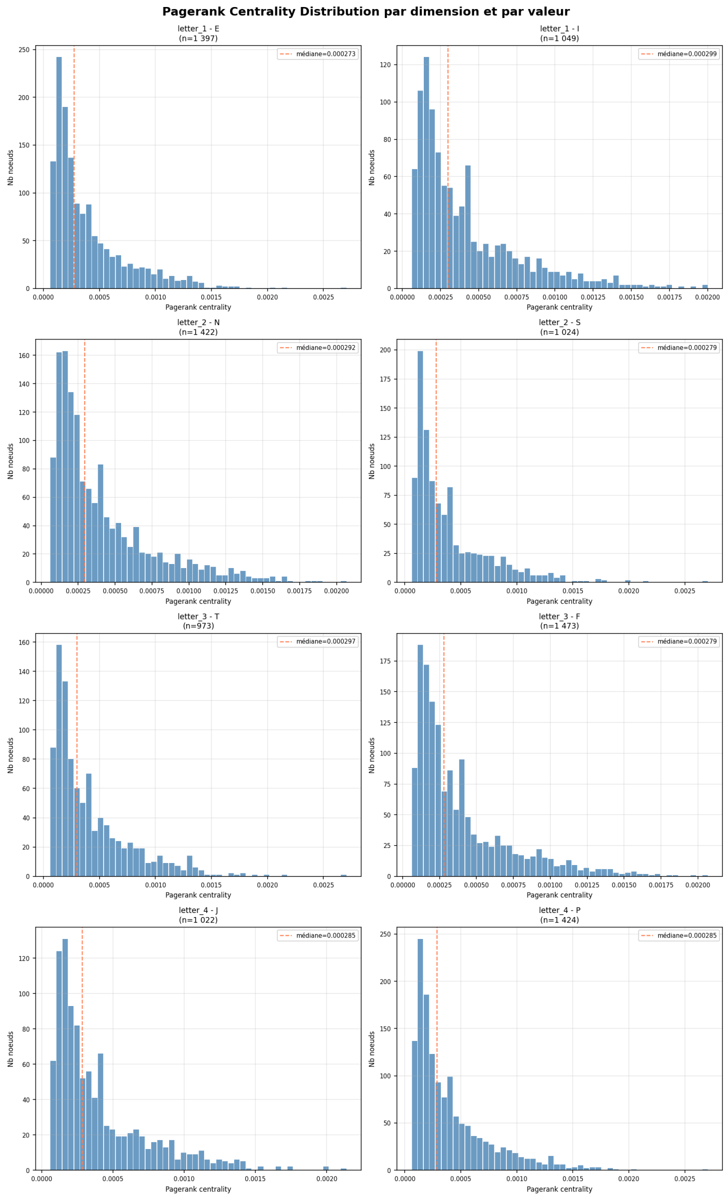

Figure 31: PageRank distribution (filtered network)

The PageRank distribution exhibits an intermediate shape between those observed for middle and near centrality. We find the initial peak characteristic of heavy-tailed distributions, but with a more gradual decrease and a less extreme tail. The medians are remarkably concentrated around 0.00028-0.00029.

Types `I, N, T` consistently display PageRank medians slightly higher than their opposites, unlike the fourth letter (`J, P`), which have a similar median. This pattern suggests that these letters tend to be connected to slightly better-connected neighbors, resulting in a marginally higher PageRank. Although this difference is more pronounced so far, it remains very small and does not allow us to conclude that there is a significant structural difference related to the MBTI profile.

3.1.2.6 Eigenvector Centrality

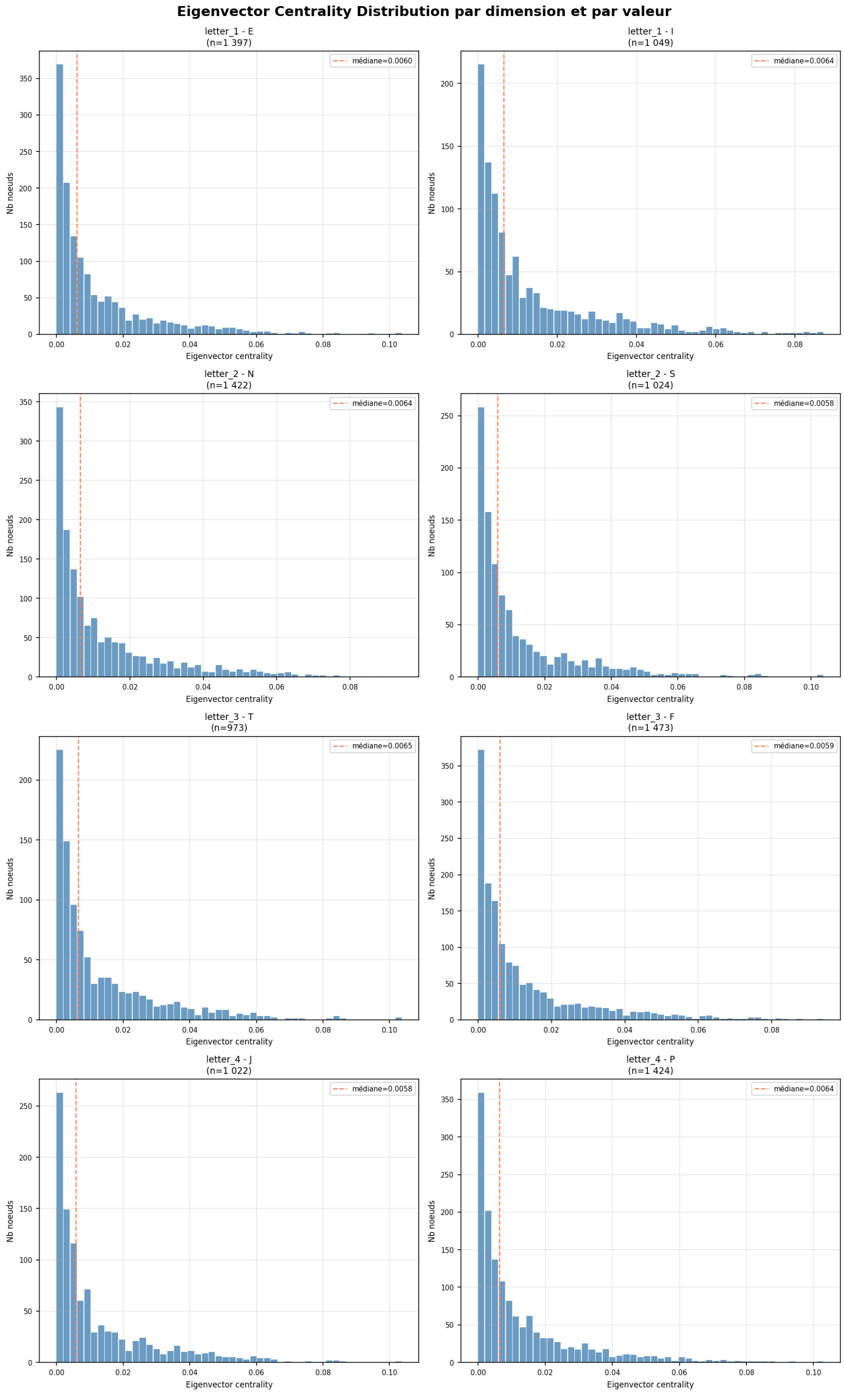

Figure 32: Eigenvector distribution (filtered network)

The distribution of eigenvector centrality has a shape similar to that of PageRank. The medians are around 0.006. The same marginal trend is observed: types `I`, `N`, `T` and `P` have medians slightly higher than their opposites. As with PageRank, this difference is small and does not allow us to conclude that there is a significant structural difference related to the MBTI profile.

3.1.2.7 Clustering Coefficient

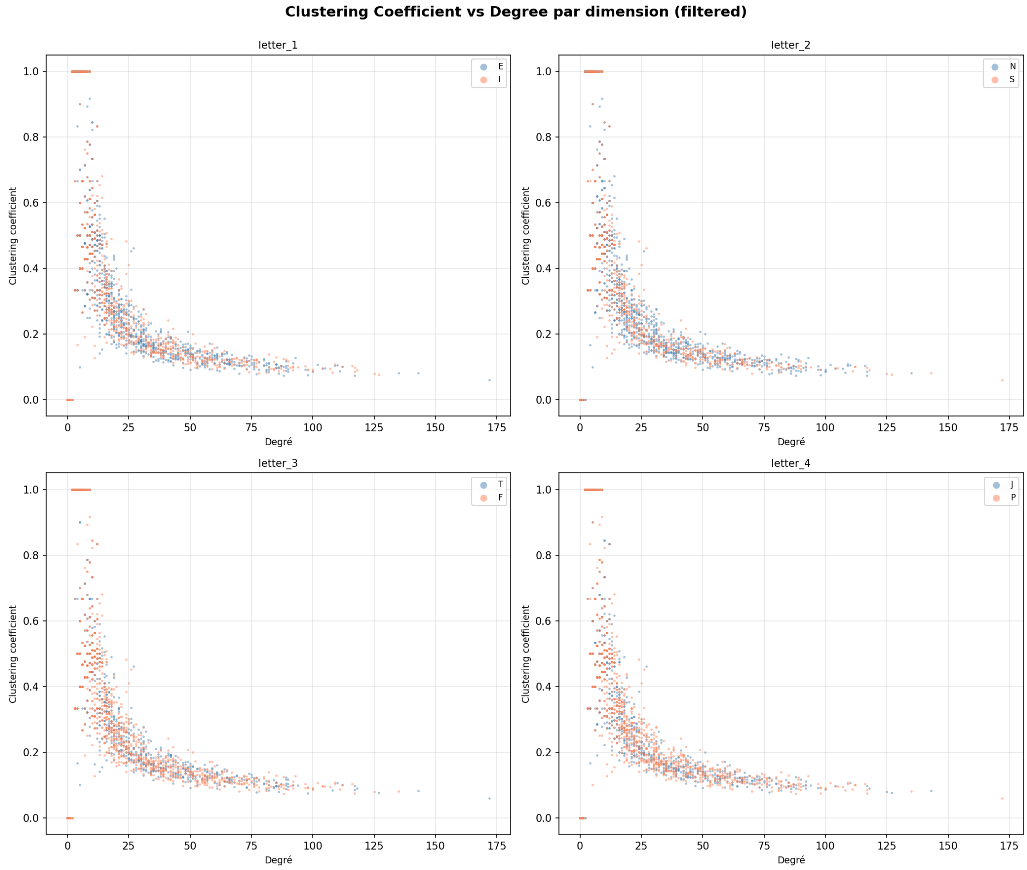

Figure 33: Clustering vs. degree (filtered network)

The observed relationship is identical across all dimensions: a clear hyperbolic decrease in the clustering coefficient as the degree increases, consistent with the expected behavior in small-world collaborative networks and with our results using the complete network.

The decrease curve between the two letters of each dimension blends indistinguishably across the entire range of degrees. This result provides the most direct and visual confirmation of the independence between MBTI profile and network structure. Actors with low degrees exhibit highly variable clustering coefficients, ranging from 0 to 1. This variability is mechanically explained: an actor who has appeared in only one film with two partners has a clustering coefficient of 1 if those two partners also worked together, and 0 otherwise.

As the degree increases, the coefficient converges to stable values around 0.1 for degrees above 50. Actors who have collaborated with more than 75 distinct partners maintain a non-zero residual clustering, indicating that their collaborators continue to cross paths with each other at a higher-than-random frequency. This is the classic signature of a professional collaboration network.

The row of points with a coefficient exactly equal to 0 (or 1) for very low degrees corresponds to actors who have exclusively appeared in a single film, either as the only known actor in the film (or with a fully interconnected group).

3.1.2.8 Rich Club Coefficient

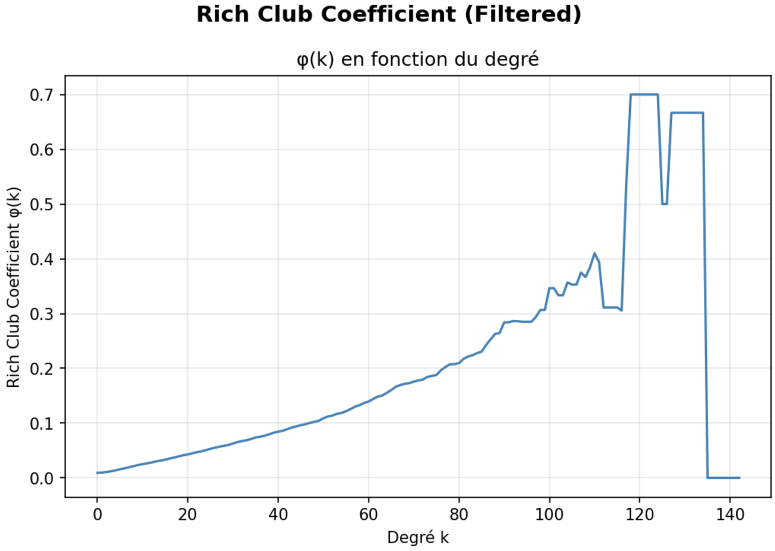

Figure 34: Rich Club vs. degree (filtered network)

The curve shows a nearly continuous increase of $\varphi(k) \approx 0.01$ for low degrees up to $\varphi(k) \approx 0.40$ around $k = 110$ before reaching a peak of $\varphi(k) \approx 0.70$ for $k \approx 120$. This monotonic progression is the signature of a rich club phenomenon: the more the analysis is restricted to the best-connected actors, the more these actors tend to have collaborated with each other.

The fluctuations observed between $k = 110$ and $k = 135$ (peak at 0.70 followed by a sharp drop to 0) are a statistical artifact related to the very small number of nodes remaining in this interval. When the subset contains only a few actors, the addition or removal of a single link causes $\varphi(k)$ to vary discontinuously. The fall to 0 for $k > 135$ indicates that there are only one or two isolated actors left at this degree level, making the coefficient not meaningfully calculable.

3.2 Heterophily

We analyze MBTI profile distributions across selected movies.

Category	Movies	Proportion (%)
Total movies	2 826	100.0 %
Movies with 2+ MBTI actors	2 417	85.53 %
Movies with 1 MBTI actor	209	7.40 %
Movies without MBTI match	200	7.08 %

Table 24: Movie distribution by MBTI match availability

3.2.1 Distribution by Movie

We can be interested in the distribution of the proportion of each letter per film:

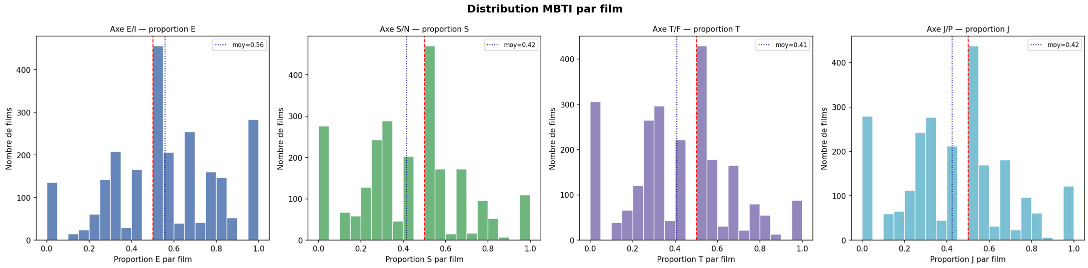

Figure 35: Distribution of each letter per movie

Our graphs exhibit a pronounced multimodal shape across all four dimensions, contrasting sharply with a unimodal distribution centered on 0.5, which would have indicated a homogeneous and systematic mix of types within the casts.

On all four axes, we observe a marked concentration at the extremes and center of the distribution, accompanied by secondary peaks. This pattern is explained by the small size of some casts: with only 2 or 3 actors matched to the MBTI, the possible proportions are necessarily discrete $(0, 0.33, 0.5, 0.67, 1)$. The axis means faithfully reproduce the bias of the source MBTI database.

Category	`E` **Dist. (%)**	`S` **Dist. (%)**	`T` **Dist. (%)**	`J` **Dist. (%)**
Actor distribution	56.9 %	41.8 %	39.9 %	41.8 %
Movie distribution	56 %	42 %	41 %	42 %

Table 25: MBTI profile distribution comparison by actor and movie

None of the four axes shows a clear concentration around any one point. The MBTI composition of a casting call results more from the random sampling of a small number of actors than from a deliberate logic of diversity or homogeneity of personality.

3.2.2 Heterophily Score

We can now look at our distribution according to our heterophilia score:

Statistic	**Value**
Movies	2 417
Mean score	0.747
Standard deviation	0.204
Minimum	0.000
1st quartile (25%)	0.656
Median (50%)	0.811
3rd quartile (75%)	0.909
Maximum	1.000

Table 26: Descriptive statistics of heterophily score

Category	Movies	Proportion (%)
Very homogeneous (score < 0.25)	60	2.5%
Mixed (0.25 ≤ score ≤ 0.75)	1 005	41.6%
Very heterogeneous (score > 0.75)	1 352	55.9%

Table 27: Movie distribution by heterophily category

The average score of 0.747, significantly higher than the theoretical median of 0.50, indicates a general trend in casting towards mixed MBTI profiles rather than homogeneous ones. More than half of the films (55.9%) have a score above 0.75, compared to only 2.5% for highly homogeneous films. This results in a very diverse distribution, demonstrating that actor groups are not correlated with their MBTI profile.

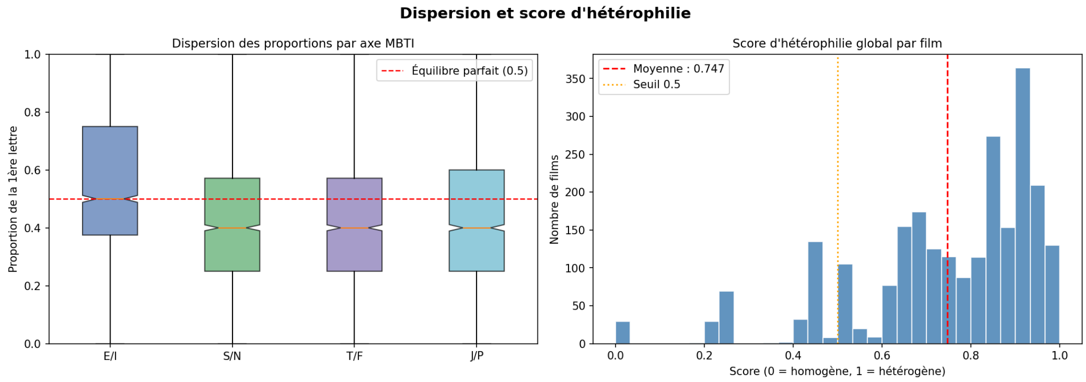

Figure 36: Boxplots and distribution of heterophily score

Box plots show almost identical dispersion across all four axes, with medians close to 0.40–0.50 and nearly symmetrical quartiles. The `E/I` axis exhibits the median closest to perfect equilibrium, while the `S/N`, `T/F` and `J/P` axes display a slightly lower median, a direct reflection of the previously identified baseline bias of the MBTI dataset. This consistency confirms that the composite heterophilia score is not pulled by any one axis but reflects a homogeneous behavior across all four dimensions.

The histogram of the heterophilia score reveals a bimodal structure: an initial peak around 0.5 and a second, broader, higher peak around 0.9. The second mode is numerically dominant, explaining the high mean. This bimodality suggests the existence of two distinct casting regimes: films with a relatively clear-cut MBTI composition on certain axes and films whose casting, generally broader, achieves an almost maximum mix across all four dimensions.

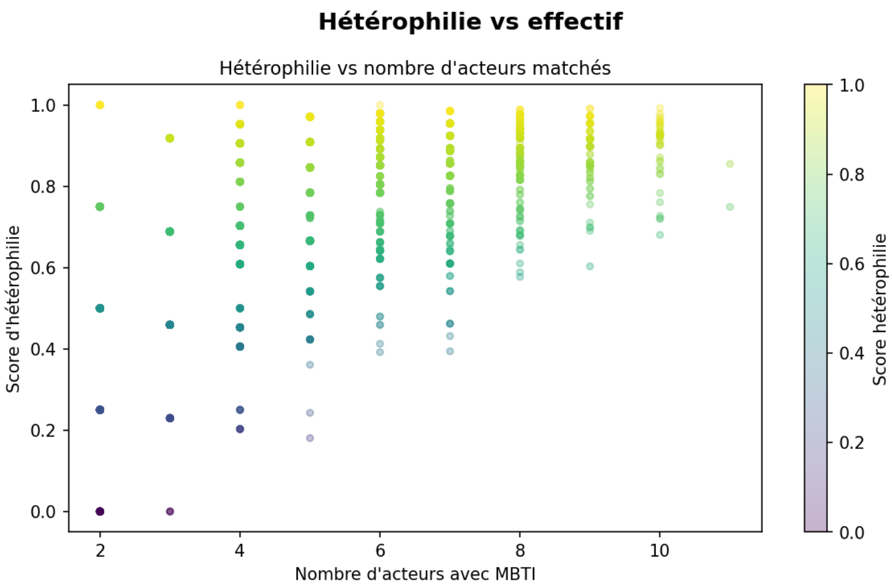

Figure 37: Heterophily score vs. actors per movie

3.2.3 Community Formation

To identify potential communities, we will assign profiles to each film. Given that several MBTI dimensions may have the same proportion in our film database, we verify that this does not constitute a majority:

Axis	Movies	Proportion (%)
E/I	456	17.36 %
S/N	470	17.90 %
T/F	429	16.34 %
J/P	438	16.68 %
At least 1 tie	1 029	39.19%

Table 28: Movies presenting proportion ties per MBTI axis

We note, however, that almost 40% of films have at least one equality in one dimension. Therefore, in the case of a equality, we will not use letters but the symbol: "=".

3.2.3.1 Movie Distribution by MBTI Profiles

Using the value "=" to replace the letter in our MBTI profile, we obtain this distribution:

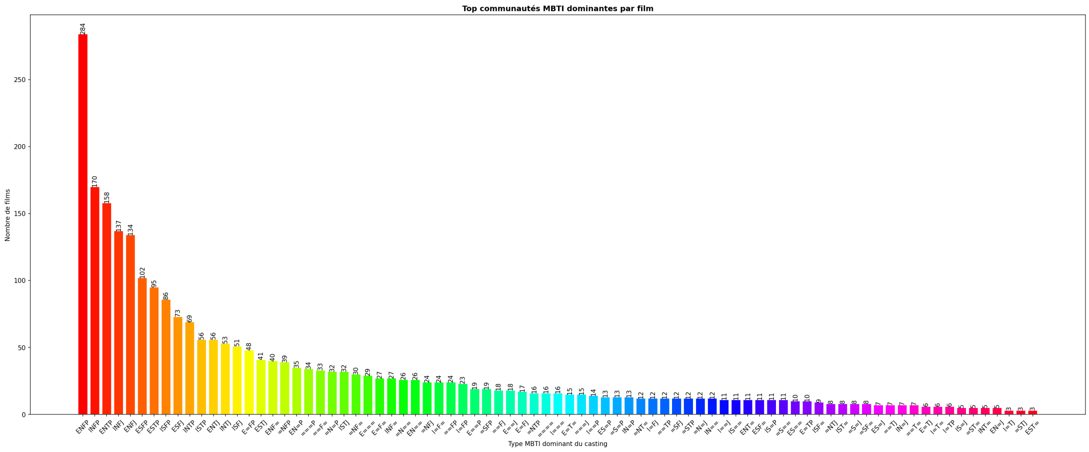

Figure 38: Movie distribution by dominant letter

We note that despite a significant number of ties, the `ENFP` profile stands out as the most frequent, occurring almost twice as often as the second most frequent. Of the ten most frequent profiles, none show a tie, yet all exhibit a clear tendency. The most frequent profile with at least one tie is `E=FP`, which also corresponds to the most frequent profile (`ENFP`).

3.2.3.2 K-means

During our calculations for the k-means, we notice that we do indeed obtain a decreasing curve and the elbow method indicates that having $k = 5$ groups would be the best compromise for our case.

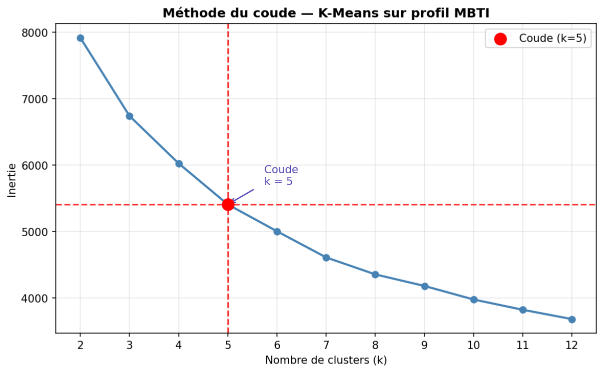

Figure 39: Elbow method on inertia as a function of the number of groups

In order to obtain relevant results, we will not simply look at the MBTI profiles but standardize them and see which profile appears more than expected in each group:

Cluster	E	S	T	J
0	-0.063	0.325	0.065	-0.192
1	0.166	0.113	-0.137	0.285
2	-0.291	-0.142	-0.197	0.094
3	0.215	-0.176	-0.052	-0.204
4	-0.072	-0.080	0.306	0.087

Table 29: Standardized average profile per cluster K-Means (k=5)

When our values are positive, they mean that the letter is more present than expected, and conversely for our negative values. The closer we get to 0, the less significant this proportion becomes. We therefore obtain these 5 profiles that appear more frequently than expected for each cluster:

Cluster	Label MBTI
0	ISTP
1	ESFJ
2	INFJ
3	ENFP
4	INTJ

Table 30: MBTI labels associated with each K-Means cluster

We can represent these trends on 2 dimensions by crossing each of our 4 initial dimensions and thus obtain 6 graphs with the centroids of our data.

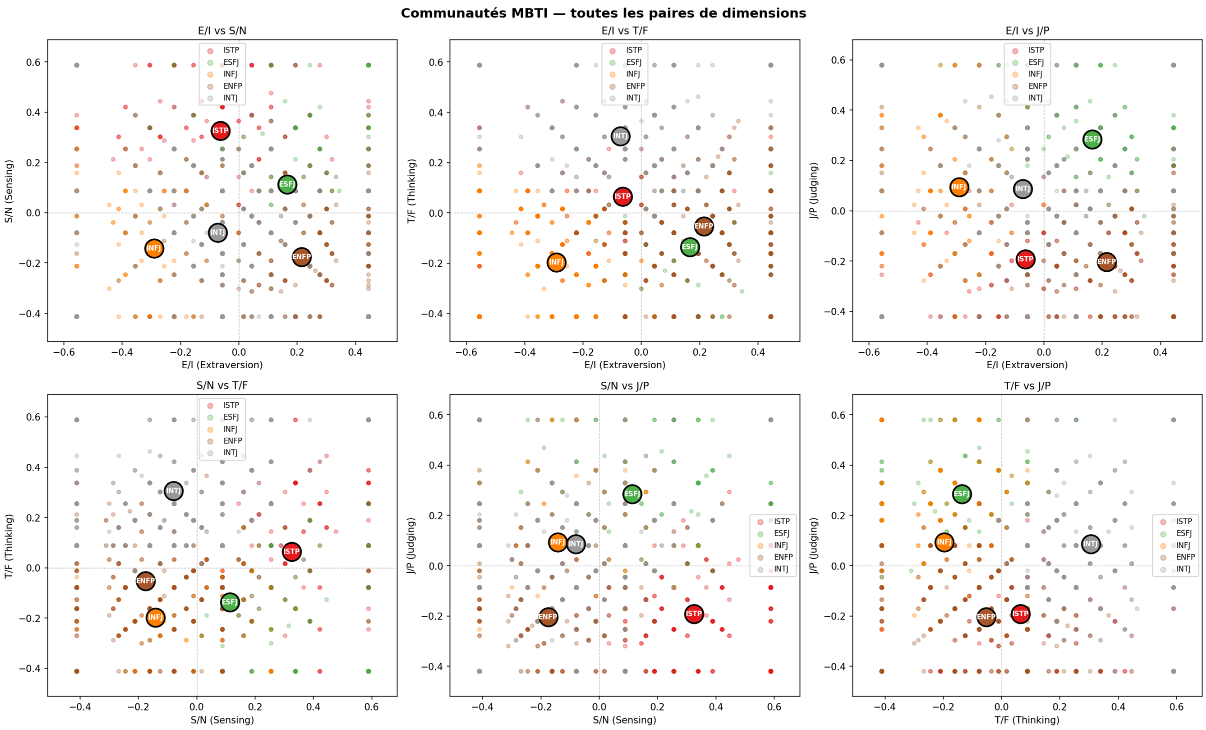

Figure 40: Distribution of groups by pair of dimension

We note that despite some scattered points, the centroids are all located around the center $(0, 0)$ in all dimensions and therefore indicate that no notable trend is present in these communities.

4 Discussion and future work

All the analyses conducted converge on a single, robust finding: MBTI personality type is not an organizing factor in collaborations between actors during the period from 1990 to 2009. The fact that all the analyses, conducted using different methodological approaches, independently point to the same conclusion reinforces the reliability of this lack of correlation.

Several hypotheses can be put forward to interpret this result. First, the film casting process is subject to commercial, artistic, and logistical constraints that, a priori, have no reason to correlate with an MBTI personality trait. Second, the MBTI data used comes from online self-reported sources, which are potentially biased in both their coverage and reliability.

The study on heterophilia complements this analysis by showing that film castings exhibit a greater mix of MBTI profiles than a homogeneous distribution would have suggested. The formation of communities using K-Means allowed for the isolation of five recurring casting profiles, but centroid analysis revealed that these groups remain close to the center of the standardized space, indicating weak structuring rather than clearly differentiated communities.

Several avenues could be explored to further develop or refine the results obtained in this project. One approach would be to extend the analysis period beyond the 1990s-2010s to verify whether the conclusions regarding the independence between MBTI profile and network structure hold true across other decades, or whether they are specific to the cinematic context of that period.

A second approach would be to analyze the personality profile using a method other than the MBTI profile to determine whether other psychological factors overlooked by the MBTI might have an influence.

Finally, it would be relevant to study finer units of analysis than the film, for example by building a network integrating the dimension of the director as a control variable, in order to verify if a link between personality and collaboration emerges or by taking into account all the actors of each film.

5 Conclusion

This project aimed to investigate the relationship between actors' MBTI personality types and the structure of film collaboration networks between 1990 and 2009, using two complementary approaches: homophily, measured by analyzing the actors' structural position within the network, and heterophily, measured by the MBTI profile composition within each film.

The homophily analysis, conducted on both the complete network and a network filtered to include only actors with known MBTI data, revealed no significant link between personality profile and an actor's structural position within the network. The transition matrices consistently displayed lift values close to 1, indicating statistical independence between the MBTI dimensions of the collaborating actors. This lack of correlation was confirmed across all seven structural metrics studied: degree, midpoint centrality, proximity centrality, PageRank, eigenvector centrality, clustering coefficient, and Rich Club coefficient. They all showed nearly identical distributions and medians between the different MBTI types, whether considering the full network or the filtered network. The only notable differences concerned the distinction between known and "unknown" nodes, the latter occupying a structurally more peripheral position.

Heterophilia analysis shed further light on the issue. The distribution of MBTI proportions per film revealed a multimodal structure strongly influenced by the small size of the matched casts, rather than a clear trend toward homogeneity or a mix of profiles. The overall heterophilia score, averaging 0.747, nevertheless revealed a statistical tendency in casting towards a mixed composition of personality profiles. This tendency turned out to be largely an artifact of the number of actors matched per film rather than a deliberate strategy of diversification during the casting process. K-Means research identified five recurring casting profiles, but these proved to be relatively undifferentiated, with their centroids remaining close to the center of the standardized space.

Ultimately, the results obtained in this project do not support the existence of a link between actors' MBTI personality type and the structure of their film collaborations. This conclusion, reached through the convergence of multiple independent analytical methods, suggests that the composition of film casts is driven more by commercial, artistic, and relational considerations than by personality affinities.

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